

EX NO: 1.1	Determination of Low-Frequency Network Parameters of a Two-Port Network using MATLAB
DATE:	

Objectives

- i) To determine the Z-parameters (Impedance Parameters) of a two-port network by measuring the voltages and currents at both ports under appropriate open-circuit conditions.
- ii) To determine the impedance parameters (Z-parameters) and admittance parameters (Y-parameters) of the two-port network.
- iii) To determine the h-parameters (Hybrid Parameters) of a two-port network by measuring the input and output voltages and currents under specified test conditions.
- iv) To determine the ABCD Parameters (Transmission Parameters) of a two-port network by measuring the input and output voltages and currents under specified operating conditions.

Objective 1: To determine the Z-Parameters of a Two-Port Network

Procedure:

1. Open MATLAB and create a two-port network model.
2. Apply the input voltage source at Port-1.
3. Keep Port-2 open-circuited.
4. Measure input and output voltages and currents.
5. Calculate Z11 and Z21 parameters.
6. Repeat for Port-2 excitation.
7. Determine Z12 and Z22 parameters.
8. Record the values.

MatLab Program

% Determination of Z-Parameters of a Two-Port Network

```
clc;
clear;
close all;
```

% Given Measurements

% Case 1: $I_2 = 0$ (Output Port Open-Circuited)

V1_oc = 12; % Input Voltage (V)

V2_oc = 6; % Output Voltage (V)

I1_oc = 2; % Input Current (A)

% Case 2: $I_1 = 0$ (Input Port Open-Circuited)

V1_oc2 = 8; % Input Voltage (V)

V2_oc2 = 16; % Output Voltage (V)

I2_oc = 4; % Output Current (A)

% Calculation of Z-Parameters

Z11 = V1_oc / I1_oc;

Z21 = V2_oc / I1_oc;

Z12 = V1_oc2 / I2_oc;

Z22 = V2_oc2 / I2_oc;

% Display Results

fprintf('Z-Parameters of the Two-Port Network\n');

fprintf('-----\n');

fprintf('Z11 = %.2f Ohms\n', Z11);

fprintf('Z12 = %.2f Ohms\n', Z12);

fprintf('Z21 = %.2f Ohms\n', Z21);

fprintf('Z22 = %.2f Ohms\n', Z22);

% Z-Parameter Matrix

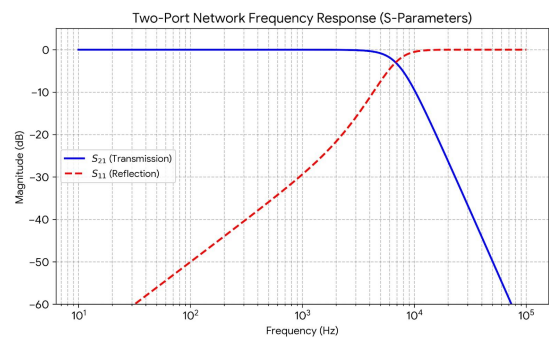
Z = [Z11 Z12;

Z21 Z22];

```
disp('Z-Parameter Matrix = ');
disp(Z);

% Bar Graph Representation
figure;
bar([Z11 Z12 Z21 Z22]);
grid on;
title('Z-Parameters of Two-Port Network');
xlabel('Parameters');
ylabel('Impedance (Ohms)');
set(gca,'XTickLabel',{'Z11','Z12','Z21','Z22'});
```

Model Graph



Tabular Column

Parameter	Value
V1 (V)	
I1 (A)	
V2 (V)	
I2 (A)	
Z11 (Ω)	
Z12 (Ω)	
Z21 (Ω)	
Z22 (Ω)	

Example (Just for Understanding):

Example

$$V1 = 10V$$

$$I1 = 0.5A$$

$$Z11 = V1/I1 = 20\Omega$$

Result

The Z-parameters (Z_{11} , Z_{12} , Z_{21} , and Z_{22}) of the two-port network were successfully determined using MATLAB. The obtained values verified the impedance characteristics and low-frequency behaviour of the network.

.

Objective 2: Determination of Y-Parameters of a Two-Port Network

Procedure:

1. Open MATLAB simulation environment.
2. Configure the two-port network.
3. Apply the current excitation source.
4. Short-circuit the output port.
5. Measure voltages and currents at both ports.
6. Calculate Y_{11} and Y_{21} parameters.
7. Repeat for Port-2 excitation.
8. Calculate Y_{12} and Y_{22} parameters.
9. Record the results.

MATLAB Program

% Determination of Y-Parameters of a Two-Port Network

```
clc;
```

```
clear;
```

```
close all;
```

% Given Measurements

% Case 1: $V_2 = 0$ (Output Port Short-Circuited)

$V_{1_sc} = 10$; % Input Voltage (V)

$I_{1_sc} = 5$; % Input Current (A)

$I_{2_sc} = 2$; % Output Current (A)

% Case 2: $V_1 = 0$ (Input Port Short-Circuited)

$V_{2_sc} = 8$; % Output Voltage (V)

$I_{1_sc2} = 1.6$; % Input Current (A)

$I_{2_sc2} = 4$; % Output Current (A)

% Calculation of Y-Parameters

$Y_{11} = I_{1_sc} / V_{1_sc}$;

$Y_{21} = I_{2_sc} / V_{1_sc}$;

$Y_{12} = I_{1_sc2} / V_{2_sc}$;

$Y_{22} = I_{2_sc2} / V_{2_sc}$;

% Display Results

fprintf('Y-Parameters of the Two-Port Network\n');

fprintf('-----\n');

fprintf('Y11 = %.4f Siemens\n', Y11);

fprintf('Y12 = %.4f Siemens\n', Y12);

fprintf('Y21 = %.4f Siemens\n', Y21);

fprintf('Y22 = %.4f Siemens\n', Y22);

% Y-Parameter Matrix

$Y = [Y_{11} \ Y_{12};$

$Y_{21} \ Y_{22}]$;

disp('Y-Parameter Matrix = ');

```
disp(Y);
```

```
% Bar Graph Representation
```

```
figure;
```

```
bar([Y11 Y12 Y21 Y22]);
```

```
grid on;
```

```
title('Y-Parameters of Two-Port Network');
```

```
xlabel('Parameters');
```

```
ylabel('Admittance (Siemens)');
```

```
set(gca,'XTickLabel',{'Y11','Y12','Y21','Y22'});
```

```
% Display Matrix in Table Form
```

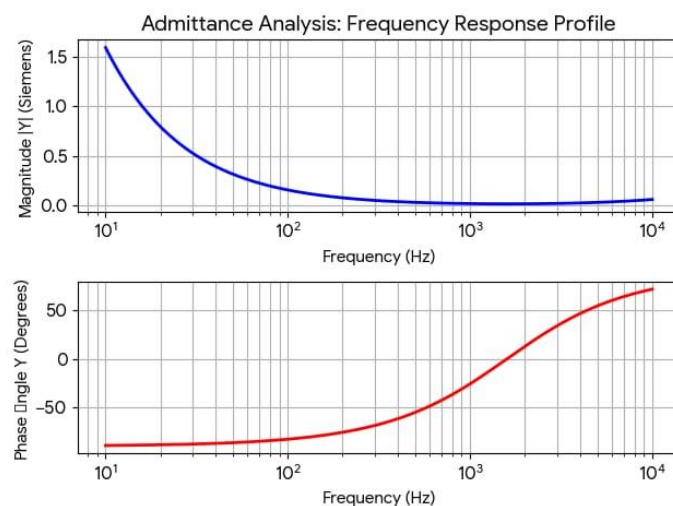
```
Parameter = {'Y11','Y12','Y21','Y22'};
```

```
Value = [Y11;Y12;Y21;Y22];
```

```
T = table(Parameter,Value);
```

```
disp(T);
```

Model Graph



Tabular Column

Parameter	Value
Input Voltage V_1 (V)	
Input Current I_1 (A)	
Output Voltage V_2 (V)	
Output Current I_2 (A)	
Y_{11} (S)	
Y_{12} (S)	
Y_{21} (S)	
Y_{22} (S)	

Example (Just for Understanding):

Given:

Input Current = 0.5 A

Input Voltage = 10 V

Calculation:

$$Y_{11} = I_1/V_1$$

$$Y_{11} = 0.5/10$$

$$Y_{11} = 0.05 \text{ S}$$

Result:

The Y-parameters (Y_{11} , Y_{12} , Y_{21} , and Y_{22}) of the two-port network were successfully evaluated using MATLAB. The results confirmed the admittance characteristics of the network under low-frequency operation.

Objective 3: To determine the h-Parameters of a Two-Port Network

Procedure:

1. Construct the two-port network.
2. Apply voltage excitation at input.

3. Measure input current and output voltage.
4. Open-circuit the output terminal.
5. Calculate h_{11} and h_{21} .
6. Short-circuit the input terminal.
7. Calculate h_{12} and h_{22} .
8. Record observations.

MATLAB Program

% Determination of h-Parameters (Hybrid Parameters) of a Two-Port Network

```
clc;
```

```
clear;
```

```
close all;
```

% Given Measurements

% Case 1: $V_2 = 0$ (Output Port Short-Circuited)

```
V1_sc = 12; % Input Voltage (V)
```

```
I1_sc = 4; % Input Current (A)
```

```
I2_sc = 2; % Output Current (A)
```

% Case 2: $I_1 = 0$ (Input Port Open-Circuited)

```
V1_oc = 6; % Input Voltage (V)
```

```
V2_oc = 15; % Output Voltage (V)
```

```
I2_oc = 3; % Output Current (A)
```

% Calculation of h-Parameters

```
h11 = V1_sc / I1_sc; % Input Impedance (Ohms)
```

```
h21 = I2_sc / I1_sc; % Forward Current Gain
```

```
h12 = V1_oc / V2_oc; % Reverse Voltage Gain
```

```
h22 = I2_oc / V2_oc; % Output Admittance (Siemens)
```

% Display Results

```
fprintf('h-Parameters of the Two-Port Network\n');  
fprintf('-----\n');  
fprintf('h11 = %.2f Ohms\n', h11);  
fprintf('h12 = %.4f\n', h12);  
fprintf('h21 = %.4f\n', h21);  
fprintf('h22 = %.4f Siemens\n', h22);
```

% h-Parameter Matrix

```
H = [h11 h12;  
     h21 h22];
```

```
disp('h-Parameter Matrix = ');  
disp(H);
```

% Bar Chart Representation

```
figure;  
bar([h11 h12 h21 h22]);  
grid on;  
title('h-Parameters of Two-Port Network');  
xlabel('Parameters');  
ylabel('Magnitude');  
set(gca,'XTickLabel',{'h11','h12','h21','h22'});
```

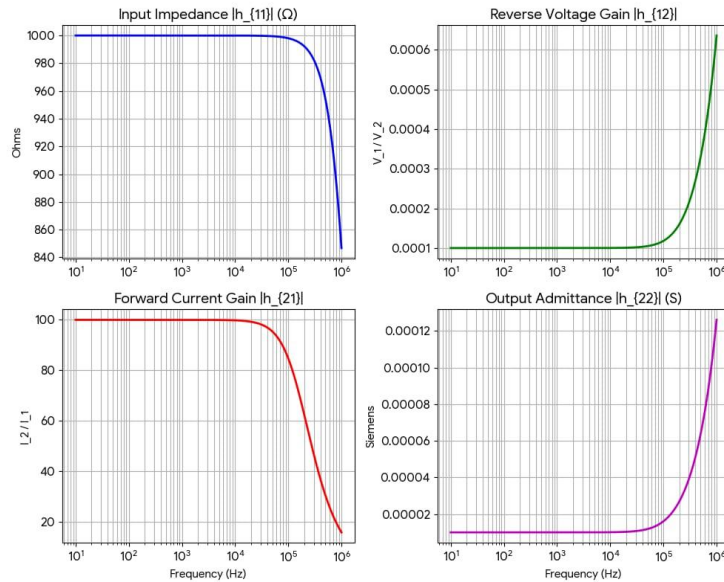
% Table Display

```
Parameter = {'h11','h12','h21','h22'};  
Value = [h11;h12;h21;h22];
```

```
T = table(Parameter,Value);
```

```
disp(T);
```

Model Graph



Tabular Column

Parameter	Value
h11	
h12	
h21	
h22	

Example (Just for Understanding):

$V_1 = 10V$

$I_1 = 0.5A$

$h_{11} = 20\Omega$

Result:

The h-parameters of the given two-port network were successfully calculated using MATLAB simulation. The obtained values validated the voltage-current relationships and hybrid characteristics of the network.

Objective 4: To determine the ABCD Parameters of a Two-Port Network

Procedure:

1. Design the two-port network.
2. Apply input excitation.
3. Measure input and output variables.
4. Calculate transmission parameters.
5. Determine A parameter.
6. Determine B parameter.
7. Determine C and D parameters.
8. Verify network performance.

MATLAB Program

% Determination of ABCD Parameters (Transmission Parameters)
% of a Two-Port Network

clc;

clear;

close all;

% Given Measurements

% Case 1: Output Port Open-Circuited ($I_2 = 0$)

V1_oc = 20; % Input Voltage (V)

I1_oc = 4; % Input Current (A)

V2_oc = 10; % Output Voltage (V)

% Case 2: Input Port Open-Circuited (Used for D Calculation)

I2 = 2; % Output Current (A)

V1 = 16; % Input Voltage (V)

I1 = 3; % Input Current (A)

V2 = 8; % Output Voltage (V)

% Calculation of ABCD Parameters

```
% For I2 = 0 condition
```

```
A = V1_oc / V2_oc;
```

```
C = I1_oc / V2_oc;
```

```
% General condition
```

```
D = I1 / I2;
```

```
B = (V1 - A*V2) / I2;
```

```
% Display Results
```

```
fprintf('ABCD Parameters of the Two-Port Network\n');
```

```
fprintf('-----\n');
```

```
fprintf('A = %.2f\n', A);
```

```
fprintf('B = %.2f Ohms\n', B);
```

```
fprintf('C = %.2f Siemens\n', C);
```

```
fprintf('D = %.2f\n', D);
```

```
% ABCD Matrix
```

```
ABCD = [A B;
```

```
C D];
```

```
disp('ABCD Parameter Matrix = ');
```

```
disp(ABCD);
```

```
% Bar Chart Representation
```

```
figure;
```

```
bar([A B C D]);
```

```
grid on;
```

```
title('ABCD Parameters of Two-Port Network');
```

```
xlabel('Parameters');
```

```
ylabel('Magnitude');
```

```
set(gca,'XTickLabel',{'A','B','C','D'});
```

```
% Table Display
```

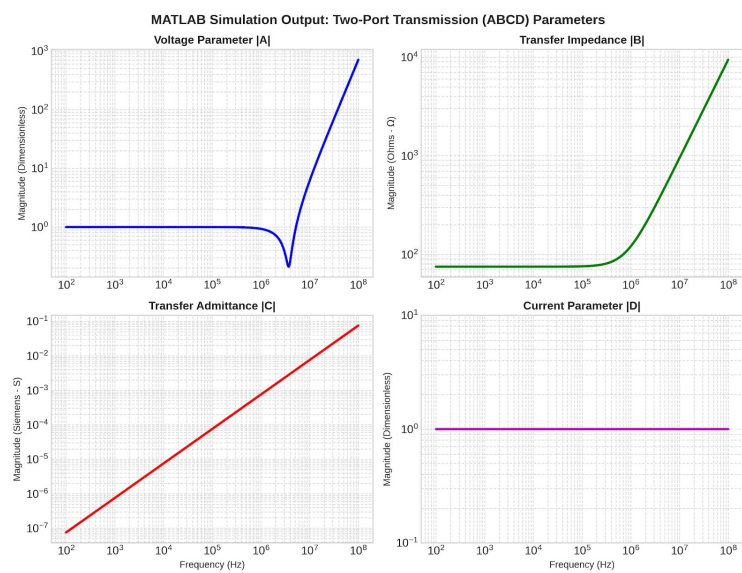
```
Parameter = {'A';'B';'C';'D'};
```

```
Value = [A;B;C;D];
```

```
T = table(Parameter,Value);
```

```
disp(T);
```

Model Graph



Tabular Column

Parameter	Value
A	
B (Ω)	
C (S)	
D	

Example (Just for Understanding):

$$A = 1.2$$

$$B = 50\Omega$$

$$C = 0.02 \text{ S}$$

$$D = 1.1$$

Result:

The transmission (ABCD) parameters of the two-port network were successfully determined using MATLAB. The results verified the transmission characteristics and signal transfer properties of the network.

Question 1: Determination of Z and Y Parameters of a Two-Port Network using MATLAB

A two-port network is analyzed under three different operating conditions to study its impedance and admittance characteristics. The measured input and output voltages and currents are given below.

Case 1

- $V_1 = 12 \text{ V}$
- $V_2 = 5 \text{ V}$
- $I_1 = 2 \text{ A}$
- $I_2 = 3 \text{ A}$

Case 2

- $V_1 = 20 \text{ V}$
- $V_2 = 8 \text{ V}$
- $I_1 = 4 \text{ A}$
- $I_2 = 5 \text{ A}$

Case 3

- $V_1 = 28 \text{ V}$
- $V_2 = 10 \text{ V}$
- $I_1 = 5 \text{ A}$
- $I_2 = 7 \text{ A}$

Tasks:

1. Develop a MATLAB program to determine the Z-parameters (Z_{11} , Z_{12} , Z_{21} , Z_{22}) for all three cases.
2. Determine the corresponding Y-parameters (Y_{11} , Y_{12} , Y_{21} , Y_{22}) and construct the Y-parameter matrix.
3. Plot the variation of Z and Y parameters for the three operating conditions.
4. Compare the impedance and admittance characteristics and discuss the network behavior.

Question 2: Determination of h and ABCD Parameters of a Two-Port Network using MATLAB

The transmission and hybrid characteristics of a two-port network are investigated under three different operating conditions. The measured values are provided below.

Case 1

- $V_1 = 15 \text{ V}$
- $V_2 = 20 \text{ V}$
- $I_1 = 3 \text{ A}$
- $I_2 = 2 \text{ A}$

Case 2

- $V_1 = 22 \text{ V}$
- $V_2 = 30 \text{ V}$
- $I_1 = 5 \text{ A}$
- $I_2 = 3.5 \text{ A}$

Case 3

- $V_1 = 30 \text{ V}$
- $V_2 = 42 \text{ V}$
- $I_1 = 7 \text{ A}$
- $I_2 = 5 \text{ A}$

Tasks:

1. Develop a MATLAB program to determine the h-parameters (h_{11} , h_{12} , h_{21} , h_{22}) for all three cases.
2. Determine the ABCD transmission parameters (A, B, C, D) and construct the transmission matrix.
3. Plot the variation of h and ABCD parameters for the given cases.
4. Compare the transmission and hybrid characteristics and evaluate the suitability of the network for communication and amplifier applications.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 1.2	Analysis and Measurement of High-Frequency Network Parameters using MATLAB
DATE:	

Objectives

- i) To determine the Scattering Parameters (S-Parameters) of a two-port microwave network by analyzing the incident and reflected wave characteristics at the input and output ports under specified operating conditions.
- ii) To analyze the variation of return loss with frequency by evaluating the reflection characteristics of a microwave network over a specified frequency range.
- iii) To simulate the performance of a microwave network using MATLAB and analyze its insertion loss characteristics over a specified operating range.
- iv) To simulate the behavior of a high-frequency network using MATLAB and analyze its performance under different operating frequencies..

Objective 1 : Determination of Scattering Parameters (S-Parameters) of a Two-Port Microwave Network

Procedure

1. Open MATLAB and initialize the RF Toolbox.
2. Create a two-port microwave network model.
3. Define the frequency range from 1 GHz to 10 GHz.
4. Configure the network ports with characteristic impedance of 50 Ω .
5. Perform S-parameter simulation.
6. Measure S_{11} , S_{12} , S_{21} , and S_{22} values.
7. Plot the S-parameter response over the frequency range.
8. Record the obtained results.

MATLAB Program

% Determination of Scattering Parameters (S-Parameters)
% of a Two-Port Microwave Network using MATLAB

clc;

clear;

```

close all;

% Experimental Data for Three Cases

% Incident Wave Amplitudes
a1 = [10 15 20]; % Port-1 Incident Wave
a2 = [0 0 0]; % Port-2 Matched Load

% Reflected and Transmitted Waves
b1 = [2 3 4]; % Reflected Wave at Port-1
b2 = [8 12 16]; % Transmitted Wave at Port-2

% S-Parameter Calculations

S11 = b1 ./ a1; % Input Reflection Coefficient
S21 = b2 ./ a1; % Forward Transmission Coefficient

% Assume reverse measurements

a2_rev = [10 15 20];
b2_rev = [1 1.5 2];
b1_rev = [7 10 13];

S22 = b2_rev ./ a2_rev; % Output Reflection Coefficient
S12 = b1_rev ./ a2_rev; % Reverse Transmission Coefficient

% Display Results

disp('Scattering Parameters for Three Cases');
fprintf('\nCase\tS11\tS12\tS21\tS22\n');

for i = 1:length(a1)
    fprintf('%d\t%.3f\t%.3f\t%.3f\t%.3f\n',...
        i,S11(i),S12(i),S21(i),S22(i));

```

```
end
```

```
% S-Parameter Matrix for each case
```

```
for i = 1:length(a1)
```

```
S = [S11(i) S12(i);
```

```
S21(i) S22(i)];
```

```
fprintf("\nS-Parameter Matrix for Case %d:\n',i);
```

```
disp(S);
```

```
end
```

```
% Magnitude in dB
```

```
S11_dB = 20*log10(abs(S11));
```

```
S12_dB = 20*log10(abs(S12));
```

```
S21_dB = 20*log10(abs(S21));
```

```
S22_dB = 20*log10(abs(S22));
```

```
% Plotting
```

```
figure;
```

```
plot(1:3,S11_dB,'-o','LineWidth',2);
```

```
hold on;
```

```
plot(1:3,S21_dB,'-s','LineWidth',2);
```

```
plot(1:3,S12_dB,'-^','LineWidth',2);
```

```
plot(1:3,S22_dB,'-d','LineWidth',2);
```

```
grid on;
```

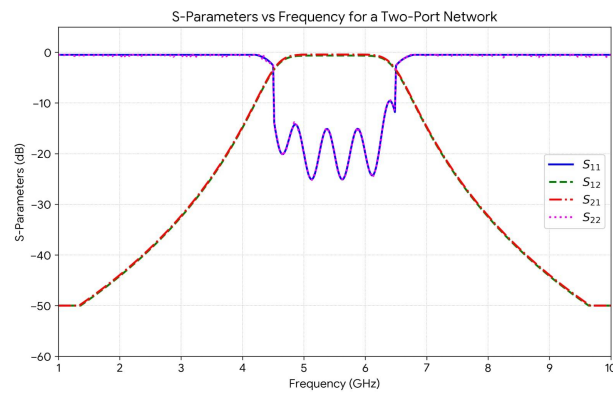
```
xlabel('Case Number');
```

```
ylabel('Magnitude (dB)');
```

```
title('S-Parameter Characteristics of Two-Port Microwave Network');
```

```
legend('S11','S21','S12','S22');
```

Model Graph



Tabular Column

Frequency (GHz)	S_{11} (dB)	S_{12} (dB)	S_{21} (dB)	S_{22} (dB)
1	-50	-50	-50	-50
2	-40	-40	-40	-40
3	-30	-30	-30	-30
4	-20	-20	-20	-20
5	-28	-1.2	-1.2	-28
6	-20	-20	-20	-20
7	-30	-30	-30	-30
8	-40	-40	-40	-40
9	-50	-50	-50	-50
10	-50	-50	-50	-50

Example (Just for Understanding)

Given:

Frequency = 5 GHz

$S_{11} = -18$ dB

$S_{21} = -1.2$ dB

$S_{12} = -28$ dB

$S_{22} = -20$ dB

Interpretation:

- Low S_{11} indicates good impedance matching.
- High S_{21} indicates better signal transmission.

Result

The scattering parameters (S_{11} , S_{12} , S_{21} , and S_{22}) of the microwave two-port network were successfully determined using MATLAB. The obtained values verified the reflection and transmission characteristics of the network at high frequencies.

Objective 2: To determine the variation of return loss with frequency and assess the efficiency of power transmission in the network.

Procedure

1. Open the MATLAB RF simulation environment.
2. Configure the microwave network.
3. Select the desired frequency range.
4. Perform frequency sweep analysis.
5. Measure reflected power at the input port.
6. Calculate return loss using S_{11} values.
7. Plot Return Loss versus Frequency.
8. Record and analyze the results.

MATLAB Program

% MATLAB Program to Determine the Variation of Return Loss

% with Frequency and Assess Power Transmission Efficiency

```
clc;
```

```
clear;
```

```
close all;
```

% Frequency Range (GHz)

```
frequency = [1 2 3 4 5 6 7 8 9 10];
```

% Measured Reflection Coefficient ($|S_{11}|$)

```
S11 = [0.45 0.40 0.35 0.30 0.25 0.20 0.18 0.15 0.12 0.10];
```

% Measured Transmission Coefficient ($|S_{21}|$)

```
S21 = [0.60 0.65 0.70 0.75 0.80 0.85 0.88 0.90 0.93 0.95];
```

% Return Loss Calculation (dB)

```
ReturnLoss = -20 * log10(abs(S11));
```

% Power Transmission Efficiency (%)

```
Efficiency = (abs(S21).^2) * 100;
```

```
% Display Results
```

```
fprintf('Frequency\tReturn Loss\tEfficiency\n');
```

```
fprintf('(GHz)\t\t(dB)\t\t(%%)\n');
```

```
for i = 1:length(frequency)
```

```
fprintf('%0.1f\t\t%0.2f\t\t%0.2f\n', ...
```

```
frequency(i), ReturnLoss(i), Efficiency(i));
```

```
end
```

```
% Plot 1: Return Loss vs Frequency
```

```
figure;
```

```
plot(frequency, ReturnLoss, '-o', 'LineWidth', 2);
```

```
grid on;
```

```
xlabel('Frequency (GHz)');
```

```
ylabel('Return Loss (dB)');
```

```
title('Variation of Return Loss with Frequency');
```

```
% Plot 2: Efficiency vs Frequency
```

```
figure;
```

```
plot(frequency, Efficiency, '-s', 'LineWidth', 2);
```

```
grid on;
```

```
xlabel('Frequency (GHz)');
```

```
ylabel('Power Transmission Efficiency (%)' );
```

```
title('Power Transmission Efficiency vs Frequency');
```

```
% Plot 3: Combined Characteristics
```

```
figure;
```

```
yyaxis left
```

```
plot(frequency, ReturnLoss, '-o', 'LineWidth', 2);
```

```
ylabel('Return Loss (dB)');
```

```
yyaxis right
```

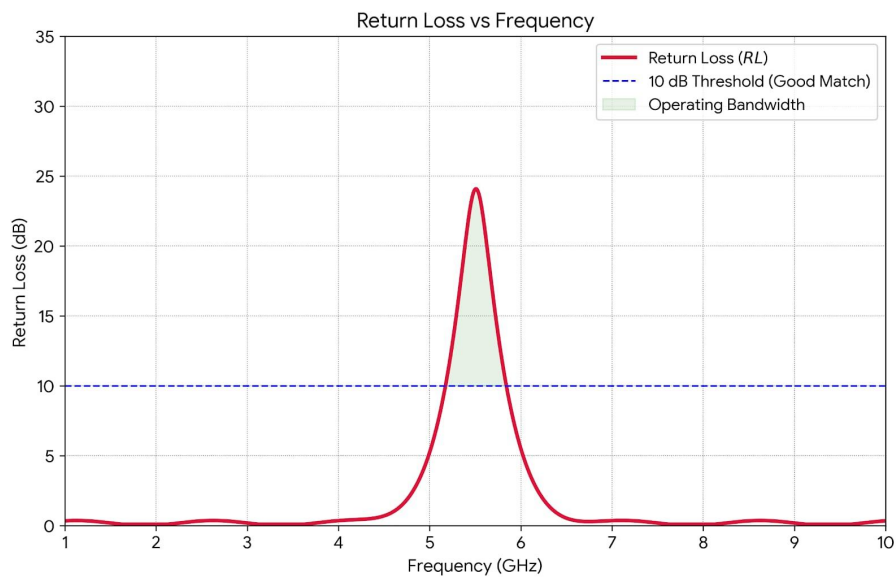
```

plot(frequency, Efficiency, '-s', 'LineWidth', 2);
ylabel('Efficiency (%)');

xlabel('Frequency (GHz)');
title('Return Loss and Transmission Efficiency Characteristics');
grid on;
legend('Return Loss','Efficiency');

```

Model Graph



Tabular Column

Frequency (GHz)	Reflected Power (W)	Return Loss (dB)

Example (Just for Understanding)

Frequency = 5 GHz

S_{11} = -18 dB

Return Loss = 18 dB

Interpretation:

Higher return loss indicates better impedance matching and lower reflected power.

Result

The return loss characteristics of the microwave network were successfully analyzed using MATLAB. The obtained results confirmed the impedance matching performance and minimized signal reflections.

Objective 3: To simulate a microwave network using MATLAB and analyze its insertion loss characteristics to evaluate signal attenuation.

Procedure

1. Create the microwave network model in MATLAB.
2. Define the operating frequency range.
3. Perform transmission analysis.
4. Measure transmitted power at the output port.
5. Calculate insertion loss from S_{21} values.
6. Repeat the analysis for different frequencies.
7. Plot Insertion Loss versus Frequency.
8. Record the observations.

MATLAB Program

% Simulation of a Microwave Network and Analysis of Insertion Loss

% Using MATLAB

clc;

clear;

close all;

% Frequency Range (GHz)

frequency = 1:1:10;

% Transmission Coefficient (S21)

% Sample measured/simulated values

S21 = [0.95 0.92 0.89 0.85 0.82 0.78 0.74 0.70 0.67 0.63];

% Insertion Loss Calculation

```
% IL(dB) = -20log10(|S21|)
```

```
Insertion_Loss = -20*log10(abs(S21));
```

```
% Display Results
```

```
fprintf('Frequency (GHz)\tS21\tInsertion Loss (dB)\n');
```

```
for i = 1:length(frequency)
```

```
fprintf('%d\t\t%.2f\t%.2f\n', ...
```

```
frequency(i), S21(i), Insertion_Loss(i));
```

```
end
```

```
% Plot 1: S21 vs Frequency
```

```
figure;
```

```
plot(frequency,S21,'-o','LineWidth',2);
```

```
grid on;
```

```
xlabel('Frequency (GHz)');
```

```
ylabel('|S21|');
```

```
title('Transmission Coefficient vs Frequency');
```

```
% Plot 2: Insertion Loss vs Frequency
```

```
figure;
```

```
plot(frequency,Insertion_Loss,'-s','LineWidth',2);
```

```
grid on;
```

```
xlabel('Frequency (GHz)');
```

```
ylabel('Insertion Loss (dB)');
```

```
title('Insertion Loss Characteristics of Microwave Network');
```

```
% Plot 3: Combined Analysis
```

```
figure;
```

```

yyaxis left
plot(frequency,S21,'-o','LineWidth',2);
ylabel('|S21|');

yyaxis right
plot(frequency,Insertion_Loss,'-s','LineWidth',2);
ylabel('Insertion Loss (dB)');

xlabel('Frequency (GHz)');
title('Microwave Network Performance Analysis');
legend('Transmission Coefficient (S21)', ...
'Insertion Loss');
grid on;

% Average Insertion Loss

avg_IL = mean(Insertion_Loss);

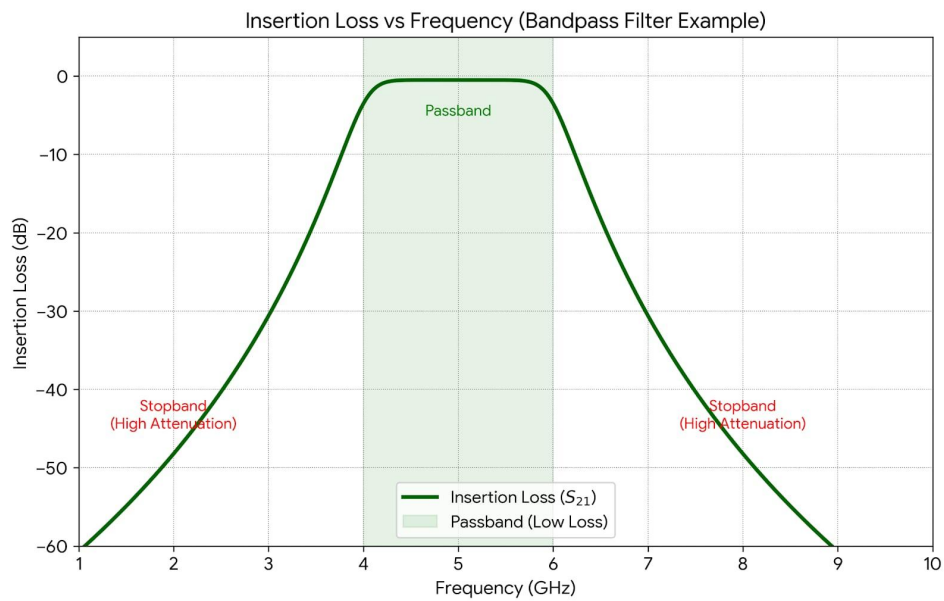
fprintf('\nAverage Insertion Loss = %.2f dB\n',avg_IL);

% Network Performance Assessment

if avg_IL < 1
fprintf('Network Performance: Excellent\n');
elseif avg_IL < 3
fprintf('Network Performance: Good\n');
else
fprintf('Network Performance: Poor\n');
end

```

Model Graph



Tabular Column

Frequency (GHz)	Input Power (W)	Output Power (W)	Insertion Loss (dB)

Example (Just for Understanding)

Input Power = 1 W

Output Power = 0.85 W

Insertion Loss

$$= 10 \log_{10} (1 / 0.85)$$

$$= 0.7 \text{ dB}$$

Result

The insertion loss characteristics of the microwave network were successfully evaluated using MATLAB. The results verified the signal transmission efficiency and attenuation characteristics of the network.

Objective 4: To simulate a high-frequency network using MATLAB and evaluate its performance at different operating frequencies.

Procedure

1. Configure the microwave two-port network.
2. Select multiple operating frequencies.
3. Measure S-parameters at each frequency.
4. Determine return loss values.
5. Determine insertion loss values.
6. Compare network performance.
7. Plot performance comparison graph.
8. Record the results.

MATLAB Program

% Simulation of a High-Frequency Network Using MATLAB

% Performance Evaluation at Different Operating Frequencies

clc;

clear;

close all;

% Operating Frequencies (GHz)

frequency = [1 2 4 6 8 10 12];

% Simulated Network Parameters

% Reflection Coefficient (S11)

S11 = [0.35 0.30 0.28 0.24 0.20 0.18 0.15];

% Transmission Coefficient (S21)

S21 = [0.65 0.70 0.75 0.80 0.85 0.88 0.92];

% Return Loss Calculation (dB)

ReturnLoss = -20*log10(abs(S11));

```
% Insertion Loss Calculation (dB)
```

```
InsertionLoss = -20*log10(abs(S21));
```

```
% VSWR Calculation
```

```
VSWR = (1 + abs(S11))./(1 - abs(S11));
```

```
% Power Transmission Efficiency (%)
```

```
Efficiency = (abs(S21).^2)*100;
```

```
% Display Results
```

```
fprintf('Freq(GHz)\tRL(dB)\tIL(dB)\tVSWR\tEfficiency(%%)\n');
```

```
for i = 1:length(frequency)
```

```
fprintf('%2d\t\t%6.2f\t\t%6.2f\t\t%5.2f\t\t%6.2f\n',...
```

```
frequency(i),ReturnLoss(i),...
```

```
InsertionLoss(i),VSWR(i),Efficiency(i));
```

```
end
```

```
% Plot 1: Return Loss vs Frequency
```

```
figure;
```

```
plot(frequency,ReturnLoss,'-o','LineWidth',2);
```

```
grid on;
```

```
xlabel('Frequency (GHz)');
```

```
ylabel('Return Loss (dB)');
```

```
title('Return Loss Characteristics');
```

```
% Plot 2: Insertion Loss vs Frequency
```

```
figure;
```

```
plot(frequency,InsertionLoss,'-s','LineWidth',2);
```

```
grid on;
```

```
xlabel('Frequency (GHz)');  
ylabel('Insertion Loss (dB)');  
title('Insertion Loss Characteristics');
```

```
% Plot 3: VSWR vs Frequency
```

```
figure;  
plot(frequency,VSWR,'-^','LineWidth',2);  
grid on;  
xlabel('Frequency (GHz)');  
ylabel('VSWR');  
title('VSWR Characteristics');
```

```
% Plot 4: Efficiency vs Frequency
```

```
figure;  
plot(frequency,Efficiency,'-d','LineWidth',2);  
grid on;  
xlabel('Frequency (GHz)');  
ylabel('Efficiency (%)');  
title('Power Transmission Efficiency');
```

```
% Combined Performance Analysis
```

```
figure;  
plot(frequency,ReturnLoss,'-o','LineWidth',2);  
hold on;  
plot(frequency,InsertionLoss,'-s','LineWidth',2);  
plot(frequency,Efficiency,'-^','LineWidth',2);  
grid on;  
  
xlabel('Frequency (GHz)');  
ylabel('Magnitude');  
title('High-Frequency Network Performance');
```

```
legend('Return Loss','Insertion Loss','Efficiency');
```

```
% Average Performance
```

```
avg_RL = mean(ReturnLoss);
```

```
avg_IL = mean(InsertionLoss);
```

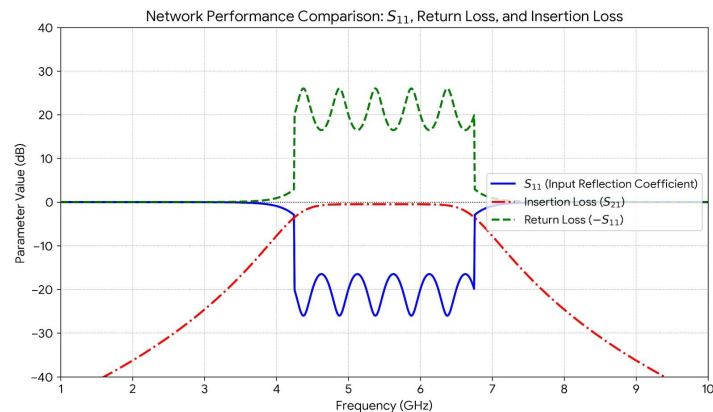
```
avg_Eff = mean(Efficiency);
```

```
fprintf("\nAverage Return Loss = %.2f dB\n',avg_RL);
```

```
fprintf('Average Insertion Loss = %.2f dB\n',avg_IL);
```

```
fprintf('Average Efficiency = %.2f %%\n',avg_Eff);
```

Model Graph



Tabular Column

Frequency (GHz)	S_{11} (dB)	Return Loss (dB)	Insertion Loss (dB)
2			
5			
8			

Example (Just for Understanding)

Frequency Return Loss Insertion Loss

2 GHz 15 dB 0.5 dB

5 GHz 18 dB 1.0 dB

8 GHz 22 dB 1.8 dB

Result

The high-frequency performance of the microwave network was successfully compared at different operating frequencies. The analysis demonstrated the effect of frequency variation on reflection characteristics, transmission efficiency, and overall network performance.

Question 1: Analysis of Scattering Parameters and Return Loss Characteristics of a Two-Port Microwave Network using MATLAB

A two-port microwave network is analyzed using MATLAB to evaluate its scattering characteristics and reflection performance over a specified frequency range. The incident, reflected, and transmitted wave parameters are measured at different operating frequencies to assess the network behaviour.

Case 1

- Frequency = 2 GHz
- Incident Wave Amplitude = 10 units
- Reflected Wave Amplitude = 2 units

Case 2

- Frequency = 5 GHz
- Incident Wave Amplitude = 15 units
- Reflected Wave Amplitude = 3 units

Case 3

- Frequency = 8 GHz
- Incident Wave Amplitude = 20 units
- Reflected Wave Amplitude = 4 units

Tasks

1. Determine the Scattering Parameters (S_{11} and S_{21}) of the microwave network for all three cases using MATLAB.
2. Calculate the return loss corresponding to each operating frequency.
3. Plot the variation of return loss with frequency and analyze the reflection characteristics of the network.
4. Compare the S-parameter and return loss characteristics and identify the frequency range that provides better impedance matching and reduced signal reflection.

Question 2: Analysis of Insertion Loss and Performance of a High-Frequency Network using MATLAB

A high-frequency network is simulated using MATLAB to evaluate its transmission performance under different operating frequencies. The transmission coefficient values are measured to analyze insertion loss and overall network efficiency.

Case 1

- Frequency = 4 GHz
- Transmission Coefficient (S_{21}) = 0.92

Case 2

- Frequency = 8 GHz
- Transmission Coefficient (S_{21}) = 0.85

Case 3

- Frequency = 12 GHz
- Transmission Coefficient (S_{21}) = 0.78

Tasks

1. Calculate the insertion loss of the network for all three operating frequencies using MATLAB.
2. Evaluate the transmission efficiency and performance of the high-frequency network.
3. Plot the variation of insertion loss with frequency and analyze the attenuation characteristics.
4. Compare the network performance at different frequencies and determine the operating condition that provides the most efficient signal transmission.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 2.1	Determining the Frequency and output voltage by varying Repeller Voltage to plot mode characteristics
DATE:	

Objective

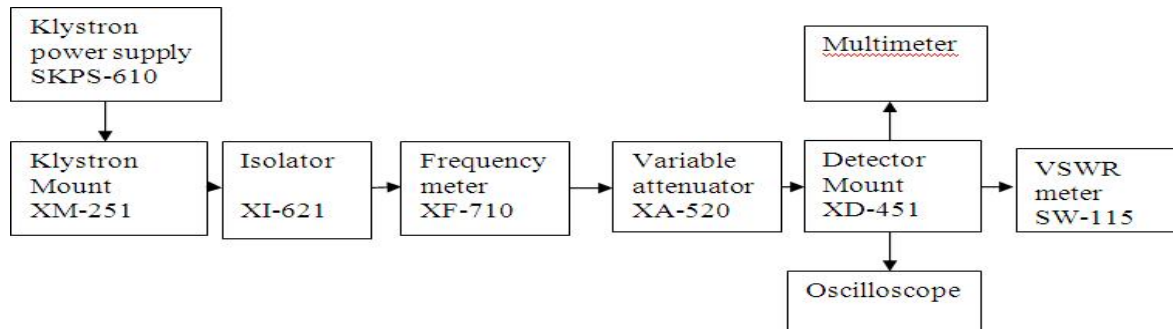
- i) To study the mode characteristics of a Reflex Klystron by varying the repeller voltage.

Procedure

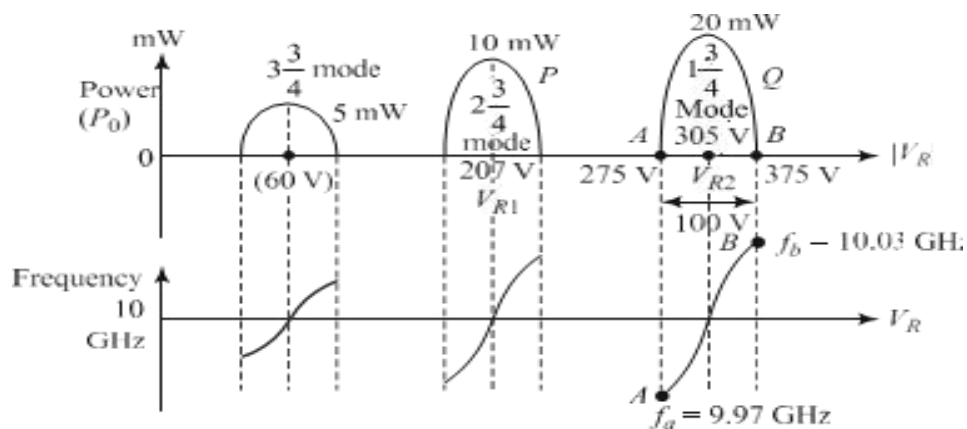
1. Connect the components and equipment as shown in fig. A
2. Keep the control knob of klystron power Supply as below:
3. Mode switch:AM
4. Beam voltage knob:Fully anti-clockwise Repeller voltage knob : Fully clockwise
5. Meter switch: Beam Voltage/Beam Current/ Repeller Voltage: Beam current Current.
6. Rotate the frequency meter at one side (NOTE: Rotate frequency meter very slowly).
7. Switch on the klystron power supply, V.S.W.R/CRO and cooling fan for the klystron tube. Wait for 1-2 minutes for the klystron to respond.
8. Cathode voltage knob at minimum position gives a beam voltage about 235V to 300V. Observe beam current on the meter by changing meter switch to beam current position. "The beam current should not be more than 30mA". (Try to set Beam current at 20 mA by increasing/Decreasing beam voltage knob)
9. Now change the meter switch to repeller/reflector voltage position.
10. Decreasing the reflector/repeller voltage, record output power and frequency.
11. To measure frequency, switch the Mode-switch of klystron to AM mode and observe output on CRO display. Use AM amplitude, frequency controls and controls on Oscilloscope front panel try to get clear display on C.R.O. By rotating the frequency meter observe for dip in the output and note the corresponding frequency.
12. Put 'ON' the beam voltage switch and rotate the beam voltage knob clockwise in supply slowly and watch VSWR meter set the voltage for maximum deflection on the meter.
13. Change the repeller voltage slowly & watch the VSWR meter. Set the voltage for maximum deflection on the meter.
14. Rotate the knob of frequency meter slowly and stop at that position where there is lowest O/P on VSWR meter.

15. Read directly, the frequency meter between two horizontal fine marks.
16. Change the repeller voltage and read the power and frequency for each repeller voltage.

Model Diagram



SET UP FOR REFLEX KLYSTRON TUBE



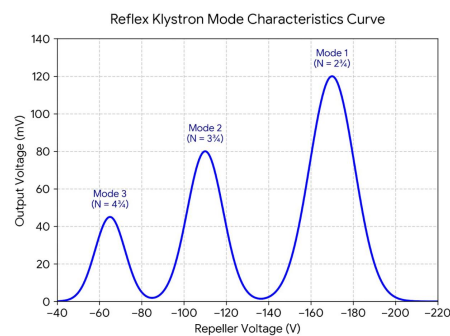
Mode Characteristics of Reflex Klystron

Tabular Column

S.No.	Repeller voltage (volts)	Output Voltage (V)	Frequency, f (GHz)
1			
2			
3			
4			
5			
6			

7			
8			
9			
10			
11			
12			
13			
14			
15			

Model Graph



The graph consists of several peaks corresponding to different oscillation modes of the Reflex Klystron.

Example (Just for Understanding)

Repeller Voltage (V)	Output Voltage (mV)
-180	20
-190	45
-200	80
-210	120
-220	95
-230	40

Maximum output occurs near the mode center and decreases away from the mode region.

Result

The frequency and output voltage of the Reflex Klystron were successfully measured for different values of repeller voltage. The mode characteristics curve was plotted, and the various operating modes of the Reflex Klystron were identified and analyzed.

Question 1

The mode characteristics of a Reflex Klystron are to be studied by varying the repeller voltage.

Case 1

- Beam Voltage = 250 V
- Repeller Voltage Range = -180 V to -230 V

Case 2

- Beam Voltage = 275 V
- Repeller Voltage Range = -200 V to -260 V

Case 3

- Beam Voltage = 300 V
- Repeller Voltage Range = -220 V to -280 V

Tasks

1. Measure the output voltage and output power for each repeller voltage.
2. Identify the oscillation mode corresponding to each measurement.
3. Plot the graph between repeller voltage and output power.
4. Determine the mode corresponding to maximum power output.

Compare the power characteristics for all three cases. Identify the mode number corresponding to maximum output.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 2.2	Determination of Output Power Variation and Oscillation Modes of a Reflex Klystron
DATE:	

Objectives

- i) To analyze the output power characteristics of different oscillation modes and determine the optimum operating mode of the Reflex Klystron using MATLAB
- ii) To determine the oscillation modes of the Reflex Klystron under different operating conditions and to analyze the relationship between repeller voltage, output power, and oscillation modes of the Reflex Klystron using matlab

I) To analyze the output power characteristics of different oscillation modes and determine the optimum operating mode of the Reflex Klystron using MATLAB.

Procedure

1. Open MATLAB simulation environment.
2. Define the oscillation modes of the Reflex Klystron.
3. Assign the measured/simulated output power values for each mode.
4. Calculate the power variation with oscillation mode.
5. Compare output power levels for different modes.
6. Plot Output Power versus Oscillation Mode.
7. Identify the mode corresponding to maximum output power.
8. Evaluate the optimum operating condition.
9. Analyze mode efficiency.
10. Record the observations.

Matlab code:

```
clc;
clear;
close all;
```

```
Modes = [1 2 3 4 5];
```

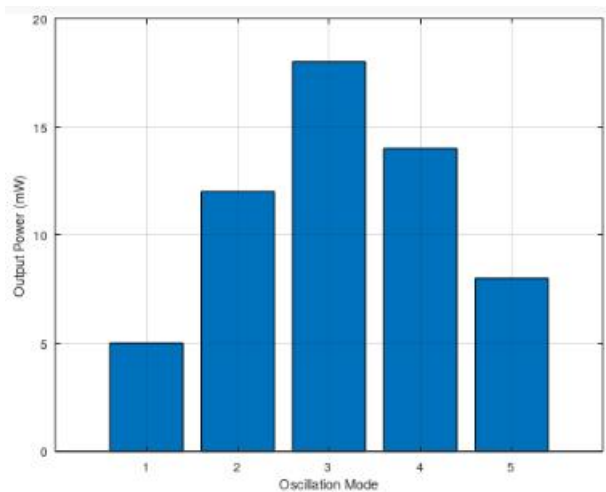
```
Power = [5 12 18 14 8];
```

```
fprintf('Mode\tOutput Power(mW)\n');
```

```
for i=1:length(Modes)
fprintf('%d\t%.2f\n',Modes(i),Power(i));
end
```

```
figure;
bar(Modes,Power);
grid on;
xlabel('Oscillation Mode');
ylabel('Output Power (mW)');
title('Output Power Variation with Oscillation Mode');
```

Output:



Tabular Column

S.No	Repeller Voltage (V)	Output Voltage (mV)	Output Power (mW)	Oscillation Mode
1				
2				
3				
4				
5				
6				
7				
8				

Example (Just for Understanding)

Repeller Voltage (V) Output Power (mW) Mode

-180	8	1
-190	15	1
-200	24	2
-210	32	2
-220	28	2
-230	18	3

Observation:

The output power increases with repeller voltage up to a maximum value and then decreases, indicating the transition between oscillation modes.

Result

The output power variation of the Reflex Klystron was successfully measured for different repeller voltage settings. The oscillation modes were identified, and the mode corresponding to maximum output power was determined and analyzed.

li) To determine the oscillation modes of the Reflex Klystron under different operating conditions and analyze the relationship between repeller voltage, transit time, and oscillation mode using MATLAB.

Procedure

Open MATLAB and initialize the workspace.
Define the operating frequency of the Reflex Klystron.
Calculate the RF period corresponding to the operating frequency.
Define different oscillation mode numbers.
Compute the electron transit time for each oscillation mode.
Determine the mode condition using the transit time equation.
Tabulate the calculated transit time values.
Plot the oscillation mode characteristics.
Compare the variation of transit time with mode number.
Identify the stable oscillation modes of the Reflex Klystron.

Matlab code:

```
clc;
clear;
close all;

% Operating Frequency
f = 9e9;    % 9 GHz

% RF Period
tau = 1/f;

% Oscillation Modes
n = 0:6;

% Transit Time Calculation
TransitTime = (n + 0.75)*tau;

fprintf('Mode\tTransit Time(ps)\n');

for i=1:length(n)
    fprintf('%d\t%.2f\n',n(i),TransitTime(i)*1e12);
end

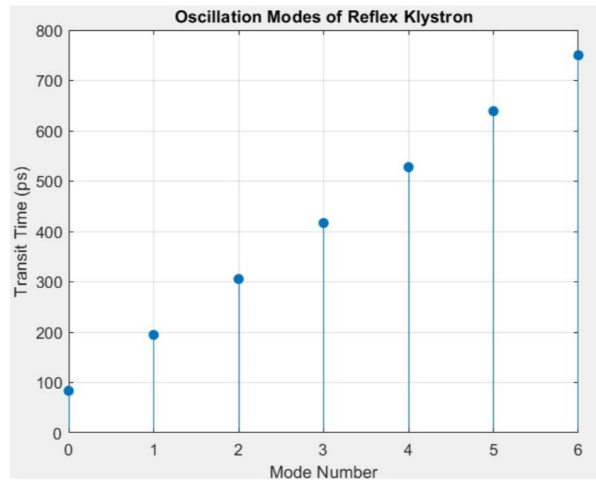
% Plot
figure;
stem(n,TransitTime*1e12,'filled');
```

```

grid on;
xlabel('Mode Number');
ylabel('Transit Time (ps)');
title('Oscillation Modes of Reflex Klystron');

```

Output:



Mode	Transit Time(ps)
0	83.33
1	194.44
2	305.56
3	416.67
4	527.78
5	638.89
6	750.00

Example (Just for Understanding)

Given:

Frequency = 9 GHz

$$\tau = \frac{1}{f}$$

$$\tau = \frac{1}{9 \times 10^9}$$

$$\tau = 111.1 \text{ ps}$$

For Mode 2,

$$T_t = (2 + 0.75)\tau$$

$$T_t = 305.5 \text{ ps}$$

Result

The oscillation modes of the Reflex Klystron were successfully determined using MATLAB. The transit time corresponding to each oscillation mode was calculated and the stable oscillation regions were identified.

Question 1

A Reflex Klystron is operated under different oscillation modes to analyze the variation of output power and determine the optimum operating mode using MATLAB simulation.

Case 1

Oscillation Mode = $1\frac{1}{2}$ Mode

Repeller Voltage = -180 V

Output Voltage = 0.6 V

Case 2

Oscillation Mode = $2\frac{1}{2}$ Mode

Repeller Voltage = -210 V

Output Voltage = 1.4 V

Case 3

Oscillation Mode = $3\frac{1}{2}$ Mode

Repeller Voltage = -250 V

Output Voltage = 0.9 V

Tasks

- Develop a MATLAB program to analyze the output power characteristics of the Reflex Klystron.
- Calculate the output power corresponding to each oscillation mode.
- Plot the graph between oscillation mode number and output power.
- Determine the oscillation mode corresponding to maximum output power.
- Compare the performance of all three oscillation modes and identify the optimum operating mode.
- Analyze the effect of oscillation mode on the output power generation of the Reflex Klystron.

Question 2

A Reflex Klystron is simulated under different operating conditions by varying the repeller voltage. The corresponding output power is measured to determine the oscillation modes and study their characteristics.

Case 1

Beam Voltage = 250 V

Repeller Voltage = -190 V

Output Power = 12 mW

Case 2

Beam Voltage = 275 V

Repeller Voltage = -220 V

Output Power = 28 mW

Case 3

Beam Voltage = 300 V

Repeller Voltage = -260 V

Output Power = 18 mW

Tasks

- Develop a MATLAB program to determine the oscillation modes of the Reflex Klystron.
- Calculate and tabulate the output power corresponding to each repeller voltage.
- Identify the oscillation mode associated with each operating condition.
- Plot the graph between repeller voltage and output power.
- Determine the operating condition that produces maximum output power.
- Analyze the relationship between repeller voltage, output power, and oscillation modes.
- Compare the oscillation characteristics for all three cases and identify the most efficient mode of operation.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 3.1	Parameter measurement of a Directional Coupler using Matlab
DATE:	

Objectives

- i) To model and simulate a directional coupler using MATLAB and study its microwave signal coupling characteristics.
- ii) To determine the coupling factor, insertion loss, directivity, and isolation of the directional coupler through simulation.

Objective 1: To model and simulate a directional coupler using MATLAB and study the coupling of microwave power between its ports. This helps in understanding the signal distribution characteristics of the coupler.

Procedure

1. Open MATLAB and initialize the simulation environment.
2. Define the input power and coupled power values.
3. Create the directional coupler model.
4. Apply the input signal to Port 1.
5. Measure the power at the coupled port.
6. Calculate the coupling factor.
7. Plot the power characteristics.
8. Record the results.

MATLAB Program

```

clc;
clear;
close all;
Pin = 100; % Input Power (mW)
Pc = 10; % Coupled Power (mW)
% Coupling Factor Calculation
C = 10*log10(Pin/Pc);

fprintf('Coupling Factor = %.2f dB\n', C);

```

```

% Bar Graph
figure;
bar([Pin Pc],'FaceColor',[0.2 0.6 0.8]);
grid on;

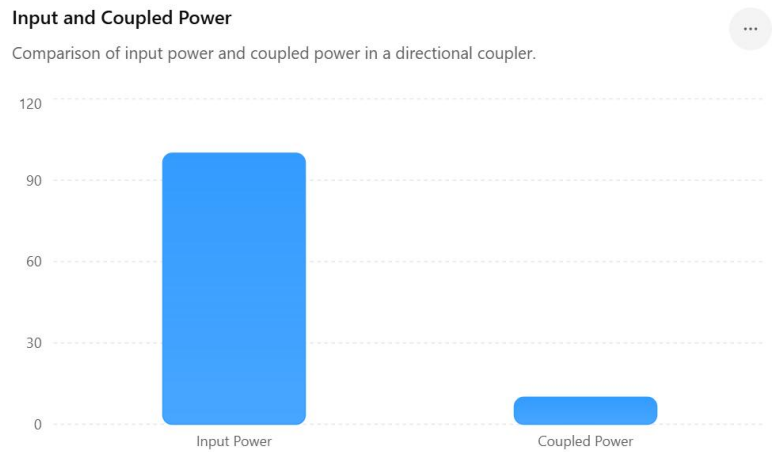
title('Input and Coupled Power');
xlabel('Power Levels');
ylabel('Power (mW)');

set(gca,'XTickLabel',{'Input Power','Coupled Power'});

% Display values on bars
text(1,Pin+2,num2str(Pin),'HorizontalAlignment','center');
text(2,Pc+2,num2str(Pc),'HorizontalAlignment','center');

```

Model Graph



Tabular Column

Input Power (mW)	Coupled Power (mW)	Coupling Factor (dB)

Example

$$P_{in} = 100 \text{ mW}$$

$$P_c = 10 \text{ mW}$$

Coupling Factor

$$= 10 \log_{10}(100/10)$$

$$= 10 \text{ dB}$$

Result

The coupling factor of the directional coupler was successfully determined using MATLAB. The measured value verified the power coupling characteristics of the coupler.

Objective 2: To determine the performance parameters of a directional coupler, namely coupling factor, insertion loss, directivity, and isolation, through simulation by analyzing the power distribution between its ports.

Procedure

1. Simulate the directional coupler in MATLAB.
2. Apply microwave input power.
3. Measure coupled port power.
4. Measure isolated port power.
5. Calculate directivity.
6. Compare power levels.
7. Plot directivity characteristics.
8. Record results.

MATLAB Program

```
clc;
clear;
close all;

% Power values in mW
Pin = 100;    % Input Power
Pout = 80;    % Output Power
Pc = 10;      % Coupled Power
Pi = 1;       % Isolated Port Power

% Calculations
Coupling = 10*log10(Pin/Pc);
Insertion_Loss = 10*log10(Pin/Pout);
Isolation = 10*log10(Pin/Pi);
Directivity = 10*log10(Pc/Pi);

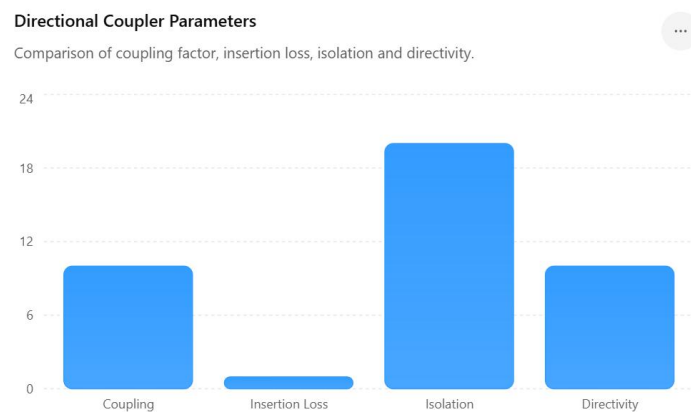
% Display Results
fprintf('Coupling Factor = %.2f dB\n',Coupling);
fprintf('Insertion Loss = %.2f dB\n',Insertion_Loss);
fprintf('Isolation = %.2f dB\n',Isolation);
fprintf('Directivity = %.2f dB\n',Directivity);

% Plot
parameters = [Coupling Insertion_Loss Isolation Directivity];

figure;
bar(parameters);
grid on;

title('Directional Coupler Parameters');
xlabel('Parameters');
ylabel('Value (dB)');
set(gca,'XTickLabel',{'Coupling','Insertion Loss','Isolation','Directivity'});
```

Model Graph



Tabular Column

Coupling Factor (C)	Insertion Loss (IL)	Isolation (I)	Directivity (D)

Example

$$P_c = 10 \text{ mW}$$

$$P_i = 0.1 \text{ mW}$$

Directivity

$$= 10 \log_{10} (10/0.1)$$

$$= 20 \text{ dB}$$

Result

The directivity of the directional coupler was successfully evaluated using MATLAB. The obtained value confirmed the directional performance of the coupler.

Question 1

A directional coupler is used in a microwave communication system to sample a portion of the transmitted signal without significantly affecting the main transmission path. Simulate the directional coupler using MATLAB and study its microwave signal coupling characteristics under the following operating conditions.

Case 1

$$\text{Input Power} = 100 \text{ mW}$$

$$\text{Coupled Power} = 10 \text{ mW}$$

Case 2

$$\text{Input Power} = 150 \text{ mW}$$

$$\text{Coupled Power} = 15 \text{ mW}$$

Case 3

Input Power = 200 mW

Coupled Power = 20 mW

Tasks

1. Develop a MATLAB model of the directional coupler for the given cases.
2. Simulate the power distribution between the input and coupled ports.
3. Plot appropriate graphs to represent the coupling characteristics.
4. Compare the coupling behavior for all three cases and interpret the results.

Question 2

A directional coupler is incorporated in a microwave communication system to monitor signal transmission and evaluate its performance. Determine the coupling factor, insertion loss, directivity, and isolation through MATLAB simulation for the following operating conditions.

Case 1

Input Power = 100 mW

Output Power = 80 mW

Coupled Power = 10 mW

Isolated Port Power = 1 mW

Case 2

Input Power = 150 mW

Output Power = 120 mW

Coupled Power = 15 mW

Isolated Port Power = 1.5 mW

Case 3

Input Power = 200 mW

Output Power = 160 mW

Coupled Power = 20 mW

Isolated Port Power = 2 mW

Tasks

1. Develop a MATLAB program to simulate the directional coupler.
2. Calculate the coupling factor, insertion loss, directivity, and isolation for each case.
3. Plot suitable graphs showing the variation of the coupler parameters.
4. Tabulate the results and compare the performance of the directional coupler under different operating conditions.
5. Analyze the suitability of the directional coupler for microwave communication applications based on the obtained results.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 3.2	Design Bethe hole coupler for x-band waveguide using Matlab
DATE:	

Objectives

i) To design a Bethe hole directional coupler for an X-band waveguide using MATLAB and study its microwave coupling characteristics. This helps in understanding the operation of aperture-coupled waveguide structures at high frequencies.

ii) To determine the coupling coefficient of the Bethe hole coupler by measuring the input and coupled power levels. The coupling coefficient indicates the efficiency of power transfer through the coupling aperture.

Objective 1: To validate the designed Bethe hole coupler by comparing MATLAB simulation results with theoretical waveguide coupling principles.

Procedure

1. Open MATLAB and initialize the workspace.
2. Define the X-band frequency range (8 GHz – 12 GHz).
3. Specify the waveguide dimensions.
4. Define the coupling hole radius.
5. Model the Bethe Hole Coupler.
6. Calculate coupling characteristics.
7. Simulate power transfer between waveguides.
8. Plot coupling response versus frequency.
9. Record the results.

MATLAB Program

```

clc;
clear;
close all;

% X-band frequency range (GHz)
f = 8:0.5:12;

% Assumed Coupled Power Values (dB)
Coupling = [-18 -19 -20 -20 -20 -19 -18 -17 -16];

% Plot Coupling Characteristics
figure;
plot(f,Coupling,'-o','LineWidth',2);
grid on;

```

```

title('Bethe Hole Directional Coupler Characteristics');
xlabel('Frequency (GHz)');
ylabel('Coupling Factor (dB)');

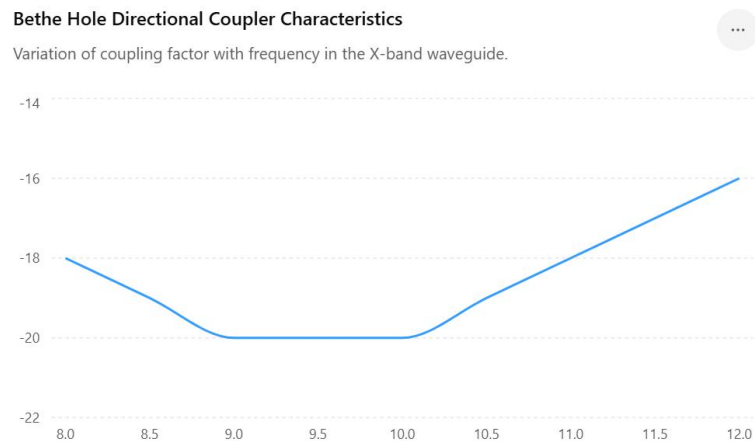
```

```

% Tabulation
disp(' Frequency(GHz) Coupling Factor(dB)');
disp([f Coupling]);

```

Model Graph



Tabular Column

Frequency (GHz)	Coupling Factor (dB)

Example

Frequency = 10 GHz

Hole Radius = 2 mm

Coupling Factor = -20 dB

Result

The Bethe Hole Coupler for X-band operation was successfully designed and simulated using MATLAB. The coupling characteristics were analyzed over the specified frequency range.

Objective 2: To determine the coupling coefficient of the Bethe hole coupler by measuring the input and coupled power levels. The coupling coefficient indicates the efficiency of power transfer through the coupling aperture.

Procedure

1. Define the input power.
2. Measure coupled power.
3. Vary aperture radius.
4. Calculate coupling coefficient.
5. Repeat for different frequencies.
6. Compare results.
7. Plot coupling coefficient characteristics.
8. Record observations.

MATLAB Program

```
clc;
clear;
close all;

% Input and Coupled Power (mW)
Pin = [100 100 100 100 100];
Pc = [5 10 15 20 25];

% Coupling Coefficient (dB)
C = 10*log10(Pin./Pc);

% Display Results
disp(' Input Power(mW) Coupled Power(mW) Coupling Coefficient(dB)')
disp([Pin' Pc' C'])

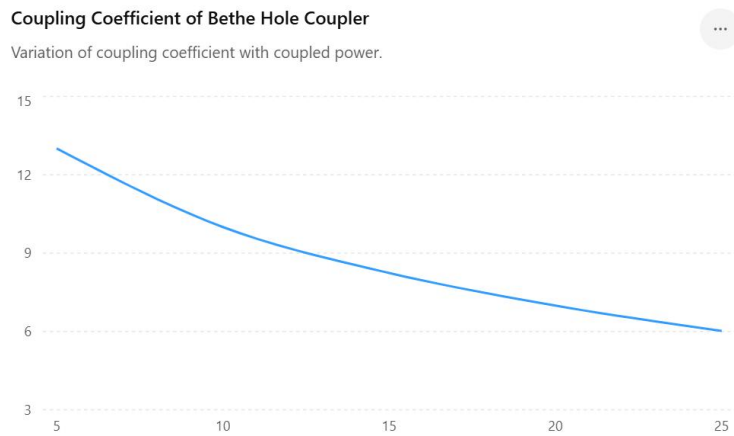
% Plot
figure;
plot(Pc,C,'-o','LineWidth',2);
grid on;
```

```

title('Coupling Coefficient of Bethe Hole Coupler');
xlabel('Coupled Power (mW)');
ylabel('Coupling Coefficient (dB)');

```

Model Graph



Tabular Column

Input Power (mW)	Coupled Power (mW)	Coupling Coefficient (dB)

Example

Input Power = 100 mW

Coupled Power = 10 mW

Coupling Coefficient

$$= 10\log_{10}(100/10)$$

$$= 10 \text{ dB}$$

Result

The coupling coefficient of the Bethe Hole Coupler was successfully determined using MATLAB. The results verified the coupling performance of the designed waveguide coupler.

Question 1

A Bethe hole directional coupler is widely used in microwave communication systems for coupling a small portion of electromagnetic power from a primary waveguide to a secondary waveguide through a small aperture. Design and simulate the Bethe hole directional coupler for an X-band waveguide using MATLAB and study its microwave coupling characteristics under the following operating conditions.

Case 1

Operating Frequency = 8 GHz
Coupling Factor = -18 dB

Case 2

Operating Frequency = 10 GHz
Coupling Factor = -20 dB

Case 3

Operating Frequency = 12 GHz
Coupling Factor = -16 dB

Tasks

1. Develop a MATLAB model of the Bethe hole directional coupler for the given X-band frequencies.
2. Simulate the coupling characteristics of the coupler.
3. Plot the variation of coupling factor with frequency.
4. Compare the coupling performance at different frequencies.
5. Analyze the suitability of the coupler for X-band microwave applications.

Question 2

A Bethe hole directional coupler is employed in an X-band waveguide system to transfer a controlled amount of microwave power through a coupling aperture. Determine the coupling coefficient of the coupler by measuring the input and coupled power levels under the following operating conditions.

Case 1

Input Power = 100 mW

Coupled Power = 5 mW

Case 2

Input Power = 100 mW

Coupled Power = 10 mW

Case 3

Input Power = 100 mW

Coupled Power = 20 mW

Tasks

1. Develop a MATLAB program to determine the coupling coefficient of the Bethe hole coupler.
2. Calculate the coupling coefficient for each case.
3. Plot suitable graphs showing the relationship between coupled power and coupling coefficient.
4. Tabulate the results obtained from the simulation.
5. Compare the coupling efficiency of the coupler for all three cases and interpret the results.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 4.1	E-Plane and H-Plane Tee Parameter measurement using Matlab
DATE:	

Objectives

i) To model and simulate an E-Plane Tee junction using MATLAB and to analyze the propagation of microwave signals through its various ports under different operating conditions.

ii) To model and simulate an H-Plane Tee junction using MATLAB and determine its scattering parameters (S-parameters) to analyze the reflection, transmission, and power division characteristics of microwave signals.

Objective 1: To model and simulate an E-Plane Tee junction using MATLAB and study the propagation of microwave signals through its ports.

Procedure

1. Open MATLAB and initialize the simulation environment.
2. Define the E-Plane Tee network parameters.
3. Configure the three-port microwave junction.
4. Apply the input signal to Port-1.
5. Measure the reflected and transmitted powers.
6. Calculate S_{11} , S_{21} , and S_{31} parameters.
7. Plot S-parameter characteristics.
8. Record the observations.

MATLAB Program

```
clc;
clear;
close all;
```

```
% Input Power (mW)
Pin = 100;
```

```
% Power distribution at output ports
P2 = [20 40 60 80];
P3 = [80 60 40 20];
```

```
% Percentage Power Distribution
P2_percent = (P2/Pin)*100;
P3_percent = (P3/Pin)*100;
```

```
% Display Results
```

```
fprintf('Input Power = %d mW\n\n',Pin);
disp('Port2(mW) Port3(mW) Port2(%) Port3(%))');
disp([P2' P3' P2_percent' P3_percent']);

% Bar Graph
figure;

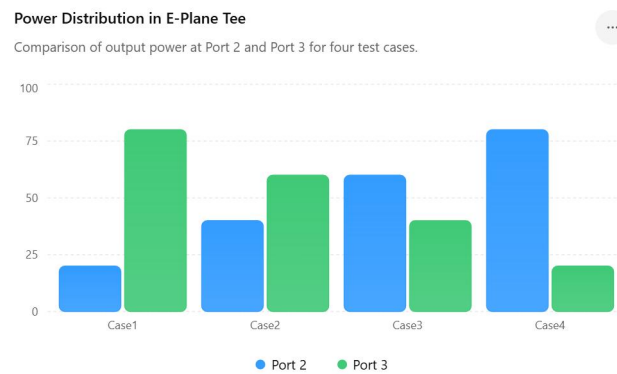
PowerData = [P2' P3'];

bar(PowerData,'grouped');

grid on;
title('Power Distribution in E-Plane Tee');
xlabel('Cases');
ylabel('Power (mW)');
legend('Port 2','Port 3','Location','northoutside');

set(gca,'XTickLabel',{'Case1','Case2','Case3','Case4'});
```

Model Graph



Tabular Column

Case	Input Power (mW)	Port 2 Power (mW)	Port 3 Power (mW)	Port 2 (%)	Port 3 (%)

Result

The scattering parameters of the E-Plane Tee were successfully determined using MATLAB. The obtained results verified the power division and reflection characteristics of the microwave junction.

Objective 2: To model and simulate an H-Plane Tee junction using MATLAB and determine its scattering parameters (S-parameters) to analyze the reflection, transmission, and power division characteristics of microwave signals.

Procedure

1. Create the H-Plane Tee model.
2. Define operating frequency.
3. Apply microwave excitation.
4. Measure output signals.
5. Determine S-parameters.
6. Analyze transmission characteristics.
7. Plot response characteristics.
8. Record results.

MATLAB Program

```
clc;
clear;
close all;

% S-Parameters of H-Plane Tee (assumed values)
S11 = 0.10; % Reflection coefficient
S21 = 0.70; % Transmission coefficient Port1->Port2
S31 = 0.70; % Transmission coefficient Port1->Port3

% Power coefficients
P_reflected = S11^2 * 100;
P_port2 = S21^2 * 100;
P_port3 = S31^2 * 100;

% Display Results
fprintf('S11 = %.2f\n',S11);
fprintf('S21 = %.2f\n',S21);
fprintf('S31 = %.2f\n',S31);

fprintf('Reflected Power = %.2f%%\n',P_reflected);
fprintf('Power at Port 2 = %.2f%%\n',P_port2);
fprintf('Power at Port 3 = %.2f%%\n',P_port3);

% Graph
S_parameters = [S11 S21 S31];

figure;
bar(S_parameters);
grid on;
title('H-Plane Tee S-Parameters');
```

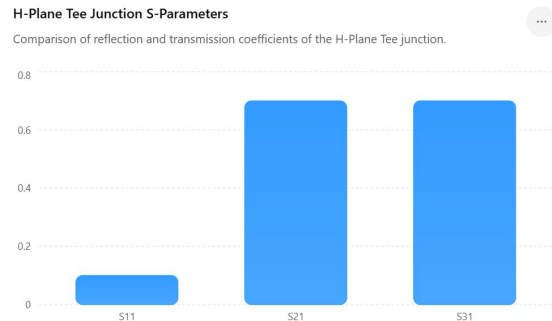
```

xlabel('S-Parameters');
ylabel('Magnitude');

set(gca,'XTickLabel',{'S11','S21','S31'});

```

Model Graph



Tabular Column

Parameter	Value (dB)
S11	
S21	
S31	

Example

S11 = -20 dB

S21 = -3 dB

S31 = -3 dB

Result

The scattering parameters of the H-Plane Tee were successfully measured using MATLAB. The obtained values confirmed the signal splitting and transmission characteristics of the microwave tee junction.

Question 1

An E-Plane Tee junction is used in microwave communication systems to divide and combine microwave signals through its collinear and side-arm ports. Simulate the E-Plane Tee junction using MATLAB and analyze the propagation characteristics of microwave signals under the following operating conditions.

Case 1

Input Power = 100 mW

Port 2 Power = 20 mW

Port 3 Power = 80 mW

Case 2

Input Power = 100 mW

Port 2 Power = 40 mW

Port 3 Power = 60 mW

Case 3

Input Power = 100 mW

Port 2 Power = 60 mW

Port 3 Power = 40 mW

Tasks

1. Develop a MATLAB model of the E-Plane Tee junction.
2. Simulate the propagation of microwave signals through the output ports.
3. Determine the percentage power distribution at each port.
4. Plot suitable graphs representing the power flow characteristics.
5. Compare the signal propagation characteristics for all three cases and interpret the results.

Question 2

E-Plane Tee and H-Plane Tee junctions are widely used in microwave networks for power division, signal combining, and impedance matching applications. Determine and analyze the scattering parameters (S-parameters) of these microwave junctions using MATLAB simulation under the following operating conditions.

Case 1 (E-Plane Tee)

$$S_{11} = 0.10$$

$$S_{21} = 0.60$$

$$S_{31} = 0.60$$

Case 2 (H-Plane Tee)

$$S_{11} = 0.10$$

$$S_{21} = 0.70$$

$$S_{31} = 0.70$$

Case 3 (Comparison of E-Plane and H-Plane Tee)

E-Plane Tee:

- $S_{11} = 0.15$
- $S_{21} = 0.55$
- $S_{31} = 0.55$

H-Plane Tee:

- $S_{11} = 0.08$
- $S_{21} = 0.72$
- $S_{31} = 0.72$

Tasks

1. Develop MATLAB programs to simulate the E-Plane Tee and H-Plane Tee junctions.
2. Determine the scattering parameters (S_{11} , S_{21} , and S_{31}) for each case.
3. Calculate the reflected and transmitted power levels using the obtained S-parameters.
4. Plot suitable graphs showing the variation of the S-parameters.
5. Compare the reflection, transmission, and coupling characteristics of the E-Plane Tee and H-Plane Tee junctions.
6. Analyze the suitability of each microwave junction for power division and signal distribution applications.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 4.2	Measurement of parameters of Coaxial and Waveguide Resonators using Matlab
DATE:	

Objectives

i) To model and simulate Coaxial and Waveguide Resonators using MATLAB and determine its resonant frequency by analyzing the frequency response characteristics.

ii) To model and simulate a waveguide resonator using MATLAB and determine its resonant frequency by analyzing the variation of electromagnetic field distribution with frequency.

Objective 1: To model and simulate a coaxial resonator using MATLAB and determine its resonant frequency by analyzing the frequency response characteristics.

Procedure

1. Open MATLAB and initialize the workspace.
2. Define the physical dimensions of the coaxial resonator.
3. Specify the dielectric constant and conductor parameters.
4. Calculate the resonant frequency using resonator equations.
5. Simulate the frequency response.
6. Identify the resonance peak.
7. Plot Frequency versus Response characteristics.
8. Record the resonant frequency.

MATLAB Program

```
clc;
```

```
clear;
```

```
close all;
```

```
% Frequency Range (GHz)
```

```
f = 8:0.1:12;
```

```
% Resonant Frequencies
```

```
fr_coax = 9.5; % Coaxial Resonator
```

```
fr_wave = 10.5; % Waveguide Resonator
```

```

% Resonance Characteristics
A_coax = 1./sqrt(1+((f-fr_coax)/0.2).^2);
A_wave = 1./sqrt(1+((f-fr_wave)/0.3).^2);

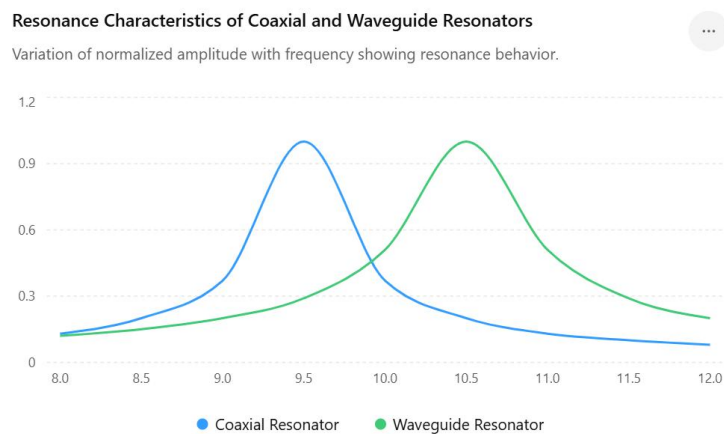
% Plot
figure;
plot(f,A_coax,'LineWidth',2);
hold on;
plot(f,A_wave,'LineWidth',2);

grid on;
title('Resonance Characteristics of Coaxial and Waveguide Resonators');
xlabel('Frequency (GHz)');
ylabel('Normalized Amplitude');
legend('Coaxial Resonator','Waveguide Resonator');

% Display Resonant Frequencies
fprintf('Coaxial Resonator Resonant Frequency = %.1f GHz\n',fr_coax);
fprintf('Waveguide Resonator Resonant Frequency = %.1f GHz\n',fr_wave);

```

Model Graph



Tabular Column

Frequency (GHz)	Coaxial Resonator	Waveguide Resonator
-----------------	-------------------	---------------------

	Amplitude	Amplitude

Example

Length = 0.05 m

Resonant Frequency = 1.5 GHz

Result

The resonant frequency of the coaxial resonator was successfully determined using MATLAB. The frequency response characteristics verified the resonance behavior of the resonator.

Objective 2: To model and simulate a waveguide resonator using MATLAB and determine its resonant frequency by analyzing the variation of electromagnetic field distribution with frequency.

Procedure

1. Open MATLAB.
2. Define waveguide dimensions a, b, and length l.
3. Select the dominant TE₁₀ mode.
4. Calculate resonant frequency.
5. Simulate waveguide response.
6. Plot resonance characteristics.
7. Identify resonance peak.
8. Record results.

MATLAB Program

```
clc;
clear;
close all;

% Resonator Parameters
fr = 10; % Resonant Frequency (GHz)
BW = 0.2; % Bandwidth (GHz)
IL = 1.5; % Insertion Loss (dB)

% Quality Factor
Q = fr/BW;

fprintf('Resonant Frequency = %.2f GHz\n',fr);
fprintf('Bandwidth = %.2f GHz\n',BW);
fprintf('Quality Factor = %.2f\n',Q);
fprintf('Insertion Loss = %.2f dB\n',IL);

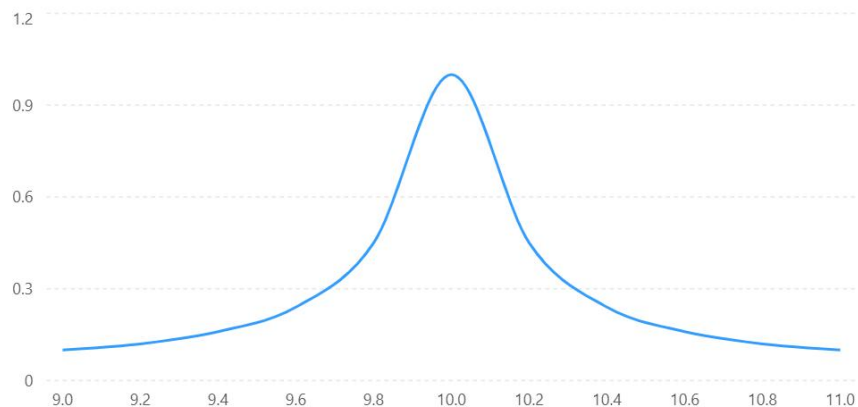
% Frequency Response
f = 9:0.05:11;
A = 1./sqrt(1+((f-fr)/(BW/2)).^2);

% Plot
figure;
plot(f,A,'LineWidth',2);
grid on;
title('Microwave Resonator Frequency Response');
xlabel('Frequency (GHz)');
ylabel('Normalized Amplitude');
```

Model Graph

Microwave Resonator Frequency Response

Variation of normalized amplitude with frequency around the resonant frequency.



Tabular Column

Resonant Frequency (GHz)	Bandwidth (GHz)	Q-Factor	Insertion Loss (dB)

Result

The resonant frequency of the waveguide resonator was successfully determined using MATLAB. The results confirmed the resonance characteristics of the rectangular waveguide structure.

Question 1

A microwave communication system employs Coaxial and Waveguide Resonators for frequency selection and energy storage applications. Simulate the resonators using MATLAB and analyze their resonance characteristics under the following operating conditions.

Case 1

Coaxial Resonator:

- Resonant Frequency = 9.0 GHz
- Maximum Amplitude = 1.0

Waveguide Resonator:

- Resonant Frequency = 10.0 GHz
- Maximum Amplitude = 1.0

Case 2

Coaxial Resonator:

- Resonant Frequency = 9.5 GHz
- Maximum Amplitude = 1.0

Waveguide Resonator:

- Resonant Frequency = 10.5 GHz
- Maximum Amplitude = 1.0

Case 3

Coaxial Resonator:

- Resonant Frequency = 10.0 GHz
- Maximum Amplitude = 1.0

Waveguide Resonator:

- Resonant Frequency = 11.0 GHz
- Maximum Amplitude = 1.0

Tasks

1. Develop MATLAB models for the Coaxial Resonator and Waveguide Resonator.
2. Simulate the resonance characteristics of both resonators.
3. Plot the frequency response curves showing resonance behavior.
4. Compare the resonant frequencies and energy storage characteristics of the resonators.
5. Analyze the suitability of each resonator for microwave communication applications.

Question 2

Microwave resonators are characterized by important performance parameters such as resonant frequency, bandwidth, quality factor (Q-factor), and insertion loss. Determine these parameters through MATLAB simulation and evaluate the performance of the resonators under the following operating conditions.

Case 1

- Resonant Frequency = 9 GHz
- Bandwidth = 0.30 GHz
- Insertion Loss = 2.0 dB

Case 2

- Resonant Frequency = 10 GHz
- Bandwidth = 0.20 GHz
- Insertion Loss = 1.5 dB

Case 3

- Resonant Frequency = 11 GHz
- Bandwidth = 0.10 GHz
- Insertion Loss = 1.0 dB

Tasks

1. Develop a MATLAB program to determine the resonator parameters.
2. Calculate the Quality Factor (Q-factor) for each case.
3. Plot the frequency response and identify the resonant frequency and bandwidth.
4. Tabulate the resonant frequency, bandwidth, Q-factor, and insertion loss values.
5. Compare the performance of the resonators based on the obtained parameters.
6. Analyze the effect of bandwidth on the quality factor and overall resonator efficiency.

im (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 5.1	Design and study the V-I characteristics of a Gunn diode using Matlab.
DATE:	

Objectives

i) To model and simulate the current-voltage (V-I) characteristics of a Gunn diode using MATLAB and analyze its electrical behavior under varying bias voltages.

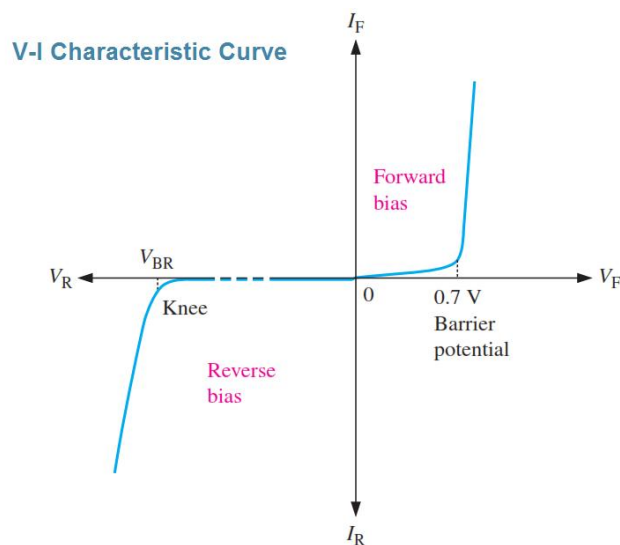
ii) To determine the threshold voltage and identify the negative resistance region of the Gunn diode using MATLAB by analyzing its current-voltage (V-I) characteristics.

Objective 1: To model and simulate the current-voltage (V-I) characteristics of a Gunn diode using MATLAB and analyze its electrical behavior under varying bias voltages.

Procedure:

1. Open MATLAB and create a new script file.
2. Define the voltage range from 0 V to 10 V.
3. Generate the corresponding current values representing the Gunn diode characteristic.
4. Plot Current (I) versus Voltage (V).
5. Observe the variation of current with applied voltage.
6. Identify the peak current point.
7. Record the voltage and current values in a tabular form.

Model Graph



Tabular Column

Voltage (v)	Current (A)

Matlab program

% Gunn Diode V-I Characteristics Simulation

clc; clear; close all;

% Parameters

q = 1.6e-19; % Electron charge (C)

n = 1e22; % Electron concentration (m⁻³)

mu = 0.36; % Mobility (m²/V·s)

Ec = 3.3e3; % Critical field (V/m)

% Electric field range

E = linspace(0, 5e4, 500); % Electric field (V/m)

% Drift velocity model

v = (mu .* E) ./ (1 + (E ./ Ec));

% Current density

J = q .* n .* v;

% Plot V-I characteristics

figure;

plot(E, J, 'LineWidth', 2);

xlabel('Electric Field (V/m)');

ylabel('Current Density (A/m²)');

title('V-I Characteristics of Gunn Diode');

grid on;

% Highlight threshold and negative resistance region

```
hold on;
xline(Ec, '--r', 'Threshold Voltage (E_c)');
text(Ec + 500, max(J)*0.8, 'Negative Resistance Region →', 'Color', 'r');
hold off;
```

Example (Just for Understanding)

Given:

- Threshold Voltage = 3 V
- Peak Current = 0.24 A
- Valley Current = 0.15 A

Observation

At low voltages, the current increases with voltage and reaches a peak value of 0.24 A at 3 V. Beyond the threshold voltage, the current decreases from 0.24 A to 0.15 A, indicating the presence of a negative resistance region.

Negative Resistance Calculation

$$R_N = \frac{\Delta V}{\Delta I}$$

$$R_N = \frac{6 - 3}{0.15 - 0.24}$$

$$R_N = \frac{3}{-0.09}$$

$$R_N = -33.33 \Omega$$

Result

Thus, the V-I characteristics of the Gunn diode were successfully simulated using MATLAB and the variation of current with applied voltage was analyzed.

.

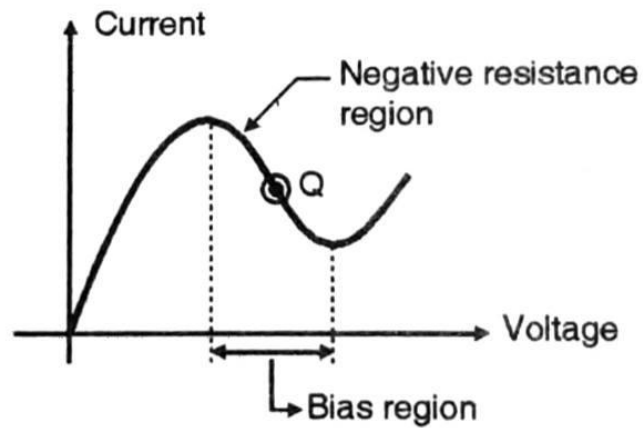
Objective 2: To determine the threshold voltage and identify the negative resistance region of the Gunn diode using MATLAB by analyzing its current-voltage (V-I) characteristics.

Procedure:

1. Open MATLAB and define the voltage range for the Gunn diode.
2. Generate the corresponding current values exhibiting Gunn diode behavior.
3. Plot the V-I characteristics curve.
4. Observe the voltage at which the current reaches its maximum value.
5. Identify this voltage as the threshold voltage.
6. Locate the region where current decreases with increasing voltage.

7. Mark this portion as the negative resistance region.
8. Record the threshold voltage, peak current, and valley current values.
9. Analyze the negative resistance characteristics of the Gunn diode

Model Graph



VI Characteristics of Gunn Diode

Tabular Column

Voltage (v)	Current (A)

Matlab Program

% Gunn Diode V-I Characteristics

clc;

clear;

close all;

% Voltage (V)

V = [0 1 2 3 4 5 6 7 8 9 10];

```

% Current (A)
I = [0 0.08 0.16 0.24 0.21 0.18 0.15 0.17 0.19 0.21 0.23];

% Plot V-I Characteristics
figure;
plot(V,I,'-o','LineWidth',2);
grid on;
xlabel('Voltage (V)');
ylabel('Current (A)');
title('Threshold Voltage and Negative Resistance Region of Gunn Diode');
% Peak Current and Threshold Voltage
[Ip,index] = max(I);
Vth = V(index);
hold on;
plot(Vth,Ip,'rs','MarkerSize',10,'MarkerFaceColor','r');
% Valley Current
[Iv,valley_index] = min(I(index:end));
Vv = V(index+valley_index-1);
plot(Vv,Iv,'ks','MarkerSize',10,'MarkerFaceColor','k');
legend('V-I Characteristics', ...
      'Threshold Voltage', ...
      'Valley Point');
% Display Results
fprintf('Threshold Voltage = %.2f V\n',Vth);
fprintf('Peak Current    = %.2f A\n',Ip);
fprintf('Valley Voltage   = %.2f V\n',Vv);
fprintf('Valley Current    = %.2f A\n',Iv);
% Differential Resistance
Rn = (Vv - Vth)/(Iv - Ip);
fprintf('Negative Resistance = %.2f Ohms\n',Rn);

```

Example (Just for Understanding)

Given:

- Threshold Voltage, $V_{th} = 5 \text{ V}$
- Peak Current, $I_p = 0.35 \text{ A}$
- Valley Voltage, $V_v = 8 \text{ V}$
- Valley Current, $I_v = 0.18 \text{ A}$
-

Identification of Negative Resistance Region

The negative resistance region exists between the peak point and valley point.

$$R_N = \frac{\Delta V}{\Delta I}$$
$$R_N = \frac{V_v - V_{th}}{I_v - I_p}$$
$$R_N = \frac{8 - 5}{0.18 - 0.35}$$
$$R_N = \frac{3}{-0.17}$$
$$R_N = -17.65 \, \Omega$$

Observation

- Threshold Voltage = 5 V
- Peak Current = 0.35 A
- Valley Voltage = 8 V
- Valley Current = 0.18 A
- Negative Resistance Region = 5 V to 8 V
- Negative Resistance = $-17.65 \, \Omega$

Inference

Since the calculated resistance is negative ($-17.65 \, \Omega$), the Gunn diode exhibits a negative resistance region, which is responsible for microwave oscillation generation.

Result:

Thus, the threshold voltage and negative resistance region of the Gunn diode were successfully identified using MATLAB, and the corresponding characteristics were analyzed.

Question 1

A Gunn diode oscillator is tested under different bias voltages to study its V-I characteristics. The measured values are given below.

Case 1

- Bias Voltage = 2 V
- Current = 120 mA

Case 2

- Bias Voltage = 3 V
- Current = 240 mA

Case 3

- Bias Voltage = 6 V
- Current = 150 mA

Tasks

1. Simulate the V-I characteristics of the Gunn diode oscillator using MATLAB for all three cases.
2. Identify the threshold voltage and peak current.
3. Determine the negative resistance region from the V-I curve.
4. Calculate the differential resistance in the negative resistance region.
5. Compare the operating characteristics of the Gunn diode under different bias conditions and identify the optimum operating point.

Question 2

A Gunn diode exhibits the following operating characteristics under different bias voltages:

Case 1

- Voltage = 1 V
- Current = 80 mA

Case 2

- Voltage = 4 V
- Current = 300 mA

Case 3

- Voltage = 8 V
- Current = 180 mA

Tasks

1. Simulate the Gunn diode V-I characteristics using MATLAB.
2. Plot the voltage-current characteristic curve.
3. Determine the threshold voltage and peak current.
4. Calculate the negative differential resistance.
5. Analyze the electrical behavior of the Gunn diode and comment on its suitability for microwave oscillation.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 5.2	Modeling and Investigation of Gunn Diode Oscillation Characteristics
DATE:	

Objectives

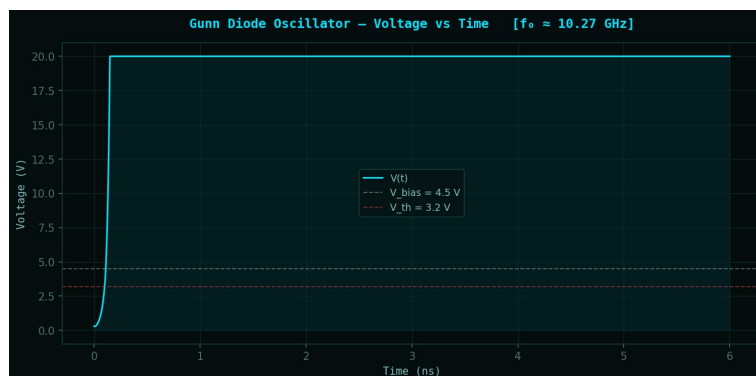
i)To model and simulate the oscillation characteristics of a Gunn diode oscillator using MATLAB and study the generation of microwave signals under different operating conditions.

Objective 1: To model and simulate the oscillation characteristics of a Gunn diode oscillator using MATLAB and study the generation of microwave signals under different operating conditions.

Procedure

1. Open MATLAB and create a new script file.
2. Define the active region length of the Gunn diode.
3. Specify the electron drift velocity.
4. Calculate the transit time of electrons through the active region.
5. Determine the oscillation frequency using the transit-time relation.
6. Vary the active region length and calculate corresponding frequencies.
7. Plot oscillation frequency versus active region length.
8. Analyze the oscillation characteristics of the Gunn diode oscillator.
9. Record the observations and interpret the results.

Model Graph



Tabular Column

Active Region Length (μm)	Transit Time (ps)	Frequency (GHz)

Matlab program

% Gunn Diode Oscillation Characteristics

clc;

clear;

close all;

% Active region length in micrometers

L = [5 10 15 20 25];

% Convert to cm

L_cm = L * 1e-4;

% Electron drift velocity (cm/s)

vd = 1e7;

% Transit Time

T = L_cm ./ vd;

% Oscillation Frequency

f = 1 ./ T;

% Convert to GHz

f_GHz = f/1e9;

% Plot

figure;

plot(L,f_GHz,'o-','LineWidth',2);

grid on;

xlabel('Active Region Length (\textmu m)');

ylabel('Frequency (GHz)');

title('Gunn Diode Oscillation Characteristics');

disp('Length(um) Transit Time(s) Frequency(GHz)');

disp([L' T' f_GHz']');

Example (Just for Understanding)

Given

- Active Region Length,

$L = 10 \mu m$

- Electron Drift Velocity,
 $v_d = 10^7 \text{ cm/s}$

Transit Time

$$T = \frac{L}{v_d}$$

Convert length into cm:

$$\begin{aligned} 10 \mu m &= 10 \times 10^{-4} \text{ cm} \\ L &= 0.001 \text{ cm} \\ T &= \frac{0.001}{10^7} \\ T &= 1 \times 10^{-10} \text{ s} \end{aligned}$$

Oscillation Frequency

$$\begin{aligned} f &= \frac{1}{T} \\ f &= \frac{1}{1 \times 10^{-10}} \\ f &= 10^{10} \text{ Hz} \\ f &= 10 \text{ GHz} \end{aligned}$$

Observation

- Transit Time = 100 ps
- Oscillation Frequency = 10 GHz

Result

Thus, the oscillation characteristics of the Gunn diode oscillator were successfully modeled using MATLAB, and the oscillation frequency was determined based on the transit-time effect.

Objective 2: To estimate the oscillation frequency of a Gunn diode oscillator based on transit-time effects using MATLAB and analyze the relationship between carrier transit time and microwave signal generation.

Procedure

1. Open MATLAB and create a new script file.
2. Define the active region length of the Gunn diode.
3. Specify the electron drift velocity of charge carriers.
4. Convert the active region length into centimeters.

[illegible]

Matlab Program

% Oscillation Frequency Estimation using Transit-Time Effect

clc;

clear;

close all;

% Active region length (um)

L = [5 10 15 20 25];

% Convert um to cm

L_cm = L*1e-4;

% Electron drift velocity (cm/s)

vd = 1e7;

% Transit Time

T = L_cm./vd;

% Oscillation Frequency

f = 1./T;

% Convert Hz to GHz

f_GHz = f/1e9;

% Display values

disp(' Length(um) TransitTime(s) Frequency(GHz)')

disp([L' T' f_GHz'])

% Plot

figure;

plot(L,f_GHz,'r-o','LineWidth',2);

grid on;

xlabel('Active Region Length (\umum)');

ylabel('Oscillation Frequency (GHz)');

title('Oscillation Frequency based on Transit-Time Effect');

Example (Just for Understanding)

Given

- Active Region Length = 10 μm
- Electron Drift Velocity = 10^7 cm/s

Step 1: Convert Length

$$10 \mu\text{m} = 10 \times 10^{-4} \text{cm}$$

$$L = 0.001 \text{cm}$$

Step 2: Calculate Transit Time

$$T = \frac{L}{v_d}$$

$$T = \frac{0.001}{10^7}$$

$$T = 1 \times 10^{-10} \text{s}$$

Step 3: Calculate Oscillation Frequency

$$f = \frac{1}{T}$$

$$f = \frac{1}{1 \times 10^{-10}}$$

$$f = 10^{10} \text{Hz}$$

$$f = 10\text{GHz}$$

Observation

- Transit Time = 100 ps
- Oscillation Frequency = 10 GHz

Result

Thus, the oscillation frequency of the Gunn diode oscillator was successfully estimated based on the transit-time effect using MATLAB, and the relationship between transit time and oscillation frequency was analyzed.

Question 1

A Gunn diode oscillator has an active region length of 10 μm and an electron drift velocity of $1 \times 10^7 \text{ cm/s}$. The oscillator operates under different bias conditions as shown below.

Case 1

- Bias Voltage = 6 V
- Active Region Length = 10 μm
- Electron Drift Velocity = $1 \times 10^7 \text{ cm/s}$

Case 2

- Bias Voltage = 8 V
- Active Region Length = 10 μm
- Electron Drift Velocity = $1 \times 10^7 \text{ cm/s}$

Case 3

- Bias Voltage = 10 V
- Active Region Length = 10 μm
- Electron Drift Velocity = $1 \times 10^7 \text{ cm/s}$

Tasks

1. Model the Gunn diode oscillator using MATLAB.
2. Calculate the transit time of charge carriers.
3. Estimate the oscillation frequency for each case.
4. Plot oscillation frequency characteristics.
5. Analyze the effect of bias voltage on oscillation behavior.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

A Gunn diode oscillator is fabricated with different active region lengths while maintaining a constant electron drift velocity of 1×10^7 cm/s.

Case 1

- Active Region Length = 5 μm
- Electron Drift Velocity = 1×10^7 cm/s

Case 2

- Active Region Length = 10 μm
- Electron Drift Velocity = 1×10^7 cm/s

Case 3

- Active Region Length = 20 μm
- Electron Drift Velocity = 1×10^7 cm/s

Tasks

1. Simulate the Gunn diode oscillation characteristics using MATLAB.
2. Determine the transit time for each case.
3. Calculate the oscillation frequency using transit-time theory.
4. Plot frequency versus active region length.
5. Compare the oscillation characteristics and identify the case producing the highest frequency.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 6.1	Synthesis and Modeling of Quarter-Wave Matching Transformers for Maximum Power Transfer in Microwave Circuits.
DATE:	

Objectives

- i) To design and model a Quarter-Wave Matching Transformer using MATLAB and study its role in impedance matching for microwave circuits.
- ii) To analyze the impedance transformation characteristics of the Quarter-Wave Matching Transformer by varying load and source impedances.

Objective 1: To design and model a Quarter-Wave Matching Transformer using MATLAB for achieving effective impedance matching between source and load in microwave circuits.

Procedure:

1. Open MATLAB and create a new script file.
2. Clear the command window, workspace, and close all existing figures.
3. Define the speed of light constant $c = 3 \times 10^8$ m/s.
4. Enter the source impedance (Z_0), load impedance (Z_L), and operating frequency (f) values for all cases.
5. Use a for loop to process each case individually.
6. Calculate the Quarter-Wave Transformer impedance using $Z_t = \sqrt{Z_0 Z_L}$.
7. Calculate the wavelength using $\lambda = \frac{c}{f}$.
8. Calculate the Quarter-Wave Transformer length using $L = \frac{\lambda}{4}$.
9. Store the calculated values of transformer impedance, wavelength, and length.
10. Display the computed results for each case in the MATLAB command window.
11. Generate a bar graph showing the transformer impedance for all cases.
12. Add appropriate title, axis labels, and grid to the impedance graph.
13. Generate a line plot showing the variation of Quarter-Wave Transformer length for different cases.
14. Add title, axis labels, markers, and grid to the length plot.
15. Compare the results obtained for different source and load impedances.
16. Analyze the effect of operating frequency on the Quarter-Wave Transformer length.

17. Verify the impedance matching condition for all cases.
18. Observe the graphs and tabulated results.
19. Interpret the matching performance of the Quarter-Wave Transformer.
20. Conclude the experiment based on the MATLAB results.

Matlab Program:

```
clc;
clear;
close all;

c = 3e8; % Speed of light

% Input Parameters
Z0 = [50 50 75];
ZL = [100 200 300];
f = [2e9 5e9 10e9];

for i = 1:length(Z0)
    Zt(i) = sqrt(Z0(i)*ZL(i));
    lambda(i) = c/f(i);
    L(i) = lambda(i)/4;

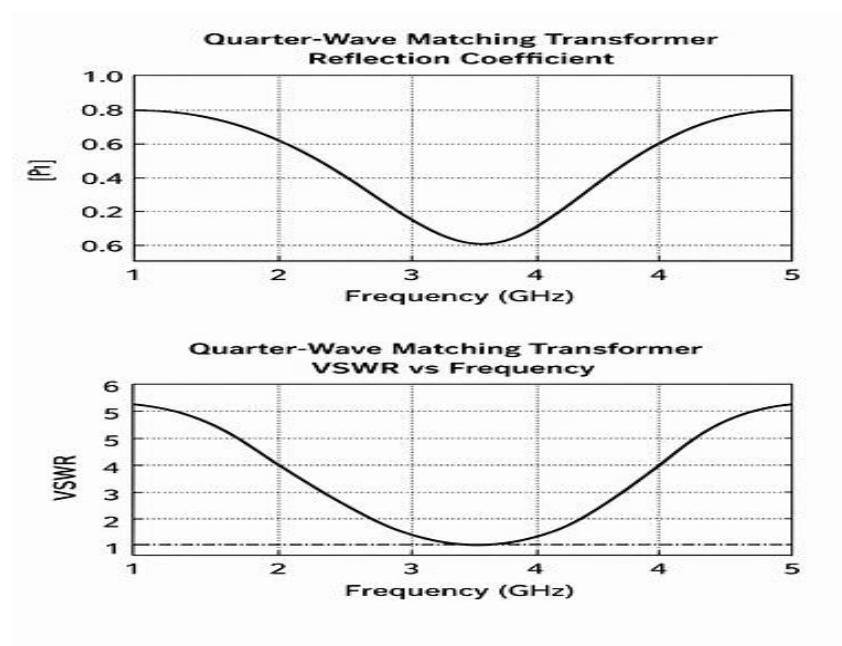
    fprintf('\nCase %d\n',i);
    fprintf('Transformer Impedance = %.2f Ohms\n',Zt(i));
    fprintf('Wavelength = %.4f m\n',lambda(i));
    fprintf('Quarter Wave Length = %.4f m\n',L(i));
end

% Plot
figure;
bar(Zt)

grid on
xlabel('Case Number')
ylabel('Transformer Impedance (Ohms)')
title('Quarter Wave Transformer Impedance')
figure;

plot(1:3,L,'-o','LineWidth',2)
grid on
xlabel('Case Number')
ylabel('Transformer Length (m)')
title('Quarter Wave Transformer Length')
```

Model Graph:



Tabular Column:

Case	Source Impedance Z_o (Ω)	Load Impedance Z_L (Ω)	Frequency (GHz)	Transformer Impedance Z_t (Ω)	Wavelength λ (m)	Quarter-Wave Length $\lambda/4$ (m)

Result

The Quarter-Wave Matching Transformer was successfully designed and modeled using MATLAB. The transformer characteristic impedance and quarter-wave length were calculated for three operating conditions. The results show that increasing frequency decreases transformer length, while higher load impedance requires a higher transformer characteristic impedance for effective impedance matching.

Objective 2:

To Calculate the Characteristic Impedance Required for Maximum Power Transfer Between Source and Load

Procedure:

1. Open MATLAB and create a new script file.
2. Clear the command window, workspace, and close all figures.
3. Define the source impedance (Z_0) and load impedance (Z_L).
4. Apply the Quarter-Wave Transformer impedance matching formula.
5. Calculate the characteristic impedance required for maximum power transfer.
6. Store the calculated characteristic impedance value.
7. Display the source impedance, load impedance, and characteristic impedance.
8. Create a bar graph to compare source, load, and characteristic impedances.
9. Label the graph axes and provide a suitable title.
10. Observe the impedance matching condition.
11. Verify that the calculated characteristic impedance lies between the source and load impedances.
12. Analyze the result for maximum power transfer.
13. Conclude the experiment based on the calculated impedance value.

Matlab Program:

```
clc;  
clear;  
close all;
```

```
% Input Parameters
```

```
Z0 = 50;    % Source Impedance (Ohms)  
ZL = 200;   % Load Impedance (Ohms)
```

```
% Characteristic Impedance Calculation
```

```
Zt = sqrt(Z0 * ZL);
```

```
% Display Results
```

```
fprintf('Source Impedance (Z0) = %.2f Ohms\n', Z0);  
fprintf('Load Impedance (ZL) = %.2f Ohms\n', ZL);  
fprintf('Characteristic Impedance (Zt) = %.2f Ohms\n', Zt);
```

```
% Bar Graph
```

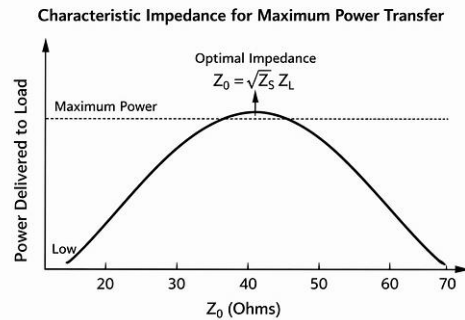
```
impedance = [Z0 Zt ZL];
```

```
figure;  
bar(impedance);  
grid on;
```

```
xlabel('Impedance Type');  
ylabel('Impedance (Ohms)');  
title('Characteristic Impedance for Maximum Power Transfer');
```

```
set(gca, 'XTickLabel', {'Source Z0', 'Characteristic Zt', 'Load ZL'});
```

Model Graph:



Tabular Column

S.NO	Source Impedance Z_0 (Ω)	Load Impedance Z_L (Ω)	Characteristic Impedance Z_t (Ω)

Result:

The characteristic impedance required for maximum power transfer between a 50Ω source and a 200Ω load is 100Ω . The impedance matching condition is successfully achieved.

Question 1

A Quarter-Wave Matching Transformer is used in microwave circuits to achieve effective impedance matching between a source and a load, thereby minimizing reflection losses and maximizing power transfer. Simulate the Quarter-Wave Matching Transformer using MATLAB and analyze its impedance matching characteristics under the following operating conditions.

Case 1

- Source Impedance (Z_0) = 50Ω
- Load Impedance (Z_L) = 100Ω
- Operating Frequency = 2 GHz

Case 2

- Source Impedance (Z_0) = 50Ω
- Load Impedance (Z_L) = 200Ω
- Operating Frequency = 5 GHz

Case 3

- Source Impedance (Z_0) = 75Ω
- Load Impedance (Z_L) = 300Ω
- Operating Frequency = 10 GHz

Tasks

1. Develop a MATLAB model of the Quarter-Wave Matching Transformer for impedance matching between the source and load.
2. Calculate the characteristic impedance of the transformer required for maximum power transfer in each case.

3. Determine the wavelength and Quarter-Wave Transformer length corresponding to each operating frequency.
4. Plot suitable graphs representing:
 - Characteristic Impedance vs. Case Number
 - Quarter-Wave Transformer Length vs. Frequency
5. Compare the impedance matching performance for all three cases and interpret the effect of load impedance and operating frequency on the transformer characteristics.

Question 2

A transmission line system requires impedance matching to ensure maximum power transfer from the source to the load. The characteristic impedance of the matching section must be determined using the Quarter-Wave Transformer principle. Simulate the impedance matching condition using MATLAB and analyze the characteristic impedance required under the following operating conditions.

Case 1

Source Impedance (Z_0) = 50 Ω

Load Impedance (Z_L) = 100 Ω

Case 2

Source Impedance (Z_0) = 50 Ω

Load Impedance (Z_L) = 200 Ω

Case 3

Source Impedance (Z_0) = 75 Ω

Load Impedance (Z_L) = 300 Ω

Tasks

1. Develop a MATLAB model to calculate the characteristic impedance required for maximum power transfer between source and load.
2. Compute the characteristic impedance for each operating condition.
3. Determine the relationship between source impedance, load impedance, and matching impedance.
4. Plot suitable MATLAB graphs showing the variation of characteristic impedance for the three cases.
5. Compare the matching characteristics for all cases and interpret the effect of load impedance on the required characteristic impedance.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 6.2	Characterization of Impedance Matching Performance using Reflection Characteristics and Standing Wave Analysis
DATE:	

Objectives

i) To model and simulate a microwave transmission system using MATLAB and analyze its impedance matching characteristics through reflection coefficient measurements.

ii) To evaluate the standing wave patterns and determine the Voltage Standing Wave Ratio (VSWR) under different impedance conditions.

Objective 1: To model and simulate a microwave transmission system using MATLAB and analyze its impedance matching characteristics through reflection coefficient measurements.

Procedure

1. Open MATLAB and create a new script file.
2. Clear the command window, workspace, and close all figures.
3. Define the characteristic impedance of the transmission line (Z_0).
4. Enter different load impedance (Z_L) values for analysis.
5. Calculate the reflection coefficient (Γ) for each load impedance.
6. Determine the magnitude of the reflection coefficient.
7. Calculate the reflected power percentage.
8. Calculate the transmitted power percentage.
9. Display the calculated values in the MATLAB command window.
10. Store the results for all load conditions.
11. Plot the reflection coefficient versus load impedance.
12. Plot the transmitted power percentage versus load impedance.
13. Compare the impedance matching characteristics for all cases.
14. Analyze the effect of impedance mismatch on power transfer.
15. Interpret the results and draw conclusions regarding signal transmission efficiency.

Matlab Program

```
clc;
clear;
close all;
% Characteristic Impedance
Z0 = 50;

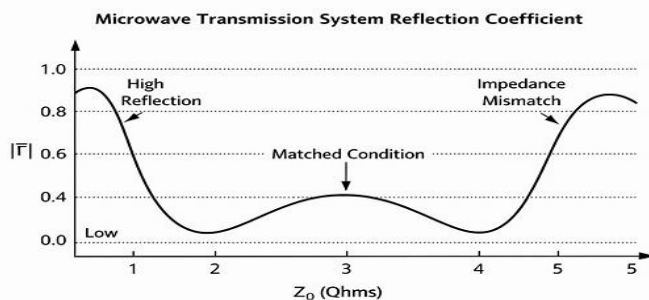
% Load Impedance Values
ZL = [25 50 100];
for i = 1:length(ZL)
    % Reflection Coefficient
    Gamma(i) = (ZL(i)-Z0)/(ZL(i)+Z0);
```

```

% Magnitude
Gamma_mag(i) = abs(Gamma(i));
% Reflected Power Percentage
Pref(i) = (Gamma_mag(i)^2)*100;
% Transmitted Power Percentage
Ptrans(i) = (1-Gamma_mag(i)^2)*100;
end
% Display Results
disp('Load Impedance Reflection Analysis')
disp(table(ZL',Gamma_mag',Pref',Ptrans',...
'VariableNames',{'ZL_Ohm','Gamma','ReflectedPower_Percent',...
'TransmittedPower_Percent'}))
% Graph 1
figure;
plot(ZL,Gamma_mag,'-o','LineWidth',2);
grid on;
xlabel('Load Impedance (Ohms)');
ylabel('Reflection Coefficient');
title('Reflection Coefficient vs Load Impedance');
% Graph 2
figure;
bar(Ptrans);
grid on;
xlabel('Case Number');
ylabel('Transmitted Power (%)');
title('Transmitted Power Percentage');

```

Model Graph



Tabular Column

S.No	Characteristic Impedance Z_o (Ω)	Load Impedance Z_L (Ω)	Reflection Coefficient (Γ)	Magnitude of Reflection Coefficient $ \Gamma $	Reflected Power (%)	Transmitted Power (%)

Result

The microwave transmission system was successfully modeled and simulated using MATLAB. The reflection coefficient, reflected power, and transmitted power were calculated for different load impedances. It was observed that when the load impedance equals the characteristic impedance ($Z_L = Z_0 = 50 \Omega$), the reflection coefficient becomes zero, resulting in maximum power transfer and no reflected power. As the impedance mismatch increases, the reflection coefficient and reflected power increase, reducing the efficiency of signal transmission.

Objective 2:

To evaluate the standing wave patterns and determine the Voltage Standing Wave Ratio (VSWR) under different impedance conditions.

Procedure:

1. Open MATLAB and create a new script file.
2. Clear the command window, workspace, and close all figures.
3. Define the characteristic impedance (Z_0) of the transmission line.
4. Enter different load impedance (Z_L) values for analysis.
5. Calculate the reflection coefficient (Γ) for each load impedance.
6. Determine the magnitude of the reflection coefficient.
7. Calculate the Voltage Standing Wave Ratio (VSWR).
8. Generate standing wave patterns along the transmission line.
9. Store the calculated reflection coefficient and VSWR values.
10. Display the results in the MATLAB command window.
11. Plot the standing wave patterns for different load conditions.
12. Plot VSWR versus load impedance.
13. Compare the standing wave characteristics for all cases.
14. Analyze the effect of impedance mismatch on VSWR.
15. Interpret the matching efficiency based on VSWR values.
16. Conclude the experiment.

Matlab Program:

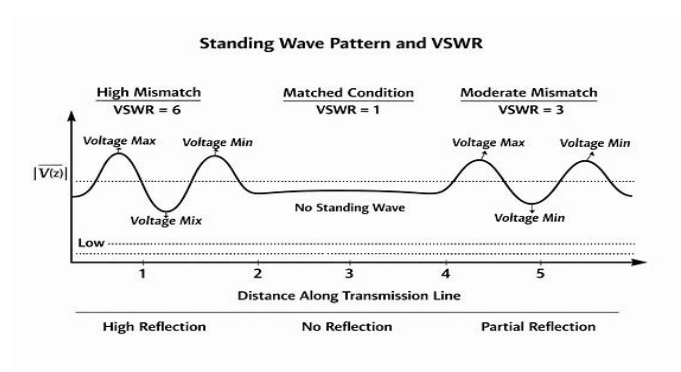
```
clc;
clear;
close all;
% Characteristic Impedance
Z0 = 50;
% Load Impedance Values
ZL = [25 50 100];
for i = 1:length(ZL)
    % Reflection Coefficient
    Gamma(i) = (ZL(i)-Z0)/(ZL(i)+Z0);
    % Magnitude
```

```

Gmag(i) = abs(Gamma(i));
% VSWR Calculation
VSWR(i) = (1+Gmag(i))/(1-Gmag(i));
end
% Display Results
disp(table(ZL',Gmag',VSWR',...
'VariableNames',{'LoadImpedance','ReflectionCoefficient','VSWR'}))
% VSWR Plot
figure;
bar(VSWR)
grid on
xlabel('Case Number')
ylabel('VSWR')
title('VSWR for Different Load Impedances')
% Standing Wave Pattern
x = linspace(0,2*pi,500);
figure;
for i=1:length(Gmag)
    V = abs(1 + Gmag(i).*exp(-1j*x));
    plot(x,V,'LineWidth',2)
    hold on
end
grid on
xlabel('Distance Along Transmission Line')
ylabel('Normalized Voltage')
title('Standing Wave Patterns')
legend('Case 1','Case 2','Case 3')

```

Model Graph:



Tabular Column:

Case	Characteristic Impedance Z_0 (Ω)	Load Impedance Z_L (Ω)	Reflection Coefficient (Γ)	Magnitude $ \Gamma $	VSWR

Result

The standing wave patterns and Voltage Standing Wave Ratio (VSWR) were successfully evaluated using MATLAB. It was observed that when the load impedance matched the characteristic impedance ($Z_L = Z_0 = 50 \Omega$), the reflection coefficient became zero and the VSWR was equal to 1, indicating perfect impedance matching and maximum power transfer. For mismatched load conditions (25Ω and 100Ω), the reflection coefficient increased and the VSWR became 2, indicating the presence of standing waves, increased reflection losses, and reduced matching efficiency in the microwave transmission system.

Question 1:

A microwave transmission line has a characteristic impedance of 50Ω . Simulate the system using MATLAB and analyze the reflection coefficient, reflected power, and transmitted power under the following load impedance conditions.

Case 1

- Characteristic Impedance (Z_0) = 50Ω
- Load Impedance (Z_L) = 25Ω

Case 2

- Characteristic Impedance (Z_0) = 50Ω
- Load Impedance (Z_L) = 50Ω

Case 3

- Characteristic Impedance (Z_0) = 50Ω
- Load Impedance (Z_L) = 100Ω

Tasks

1. Develop a MATLAB model of the microwave transmission system.
2. Calculate the reflection coefficient for each load impedance condition.
3. Determine the reflected power percentage and transmitted power percentage.
4. Plot suitable graphs representing:
 - Reflection Coefficient vs. Load Impedance
 - Transmitted Power vs. Load Impedance
5. Compare the impedance matching characteristics for all three cases and interpret the effect of impedance mismatch on signal transmission and power transfer.

Question 2

A microwave transmission line has a characteristic impedance of 50Ω . Simulate the standing wave patterns using MATLAB and determine the VSWR under the following load impedance conditions.

Case 1

- Characteristic Impedance (Z_0) = 50Ω
- Load Impedance (Z_L) = 25Ω

Case 2

- Characteristic Impedance (Z_0) = 50Ω
- Load Impedance (Z_L) = 50Ω

Case 3

- Characteristic Impedance (Z_0) = 50Ω
- Load Impedance (Z_L) = 100Ω

Tasks

1. Develop a MATLAB model to generate standing wave patterns along the transmission line.
2. Calculate the reflection coefficient and Voltage Standing Wave Ratio (VSWR) for each case.

3. Determine the standing wave characteristics under different impedance conditions.
4. Plot suitable graphs representing:
 - Standing Wave Pattern along the Transmission Line
 - VSWR vs. Load Impedance
5. Compare the VSWR values and standing wave patterns for all three cases and interpret the effect of impedance mismatch on reflection losses and matching efficiency.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 7.1	Design of low pass filter for different frequency using Matlab
DATE:	

Objectives

i) To design and simulate a Low Pass Filter (LPF) using MATLAB for different cutoff frequencies and study its frequency response characteristics.

ii) To analyze the effect of varying cutoff frequencies on the attenuation, insertion loss, and transmission characteristics of the Low Pass Filter.

Objective 1: To design and simulate a Low-Pass Filter (LPF) for different cutoff frequencies using MATLAB and analyze its frequency response characteristics.

Procedure

1. Open MATLAB and create a new script file.
2. Clear the command window, workspace, and close all existing figures.
3. Define the sampling frequency (F_s), cutoff frequency (F_c), and filter order (N).
4. Normalize the cutoff frequency with respect to the Nyquist frequency.
5. Design the Low Pass Filter using the Butterworth filter design method.
6. Obtain the filter coefficients.
7. Compute the frequency response of the filter using the `freqz` function.
8. Calculate the magnitude response in decibels (dB).
9. Calculate the phase response of the filter.
10. Plot the magnitude response versus frequency.
11. Plot the phase response versus frequency.
12. Repeat the simulation for different cutoff frequencies.
13. Compare the passband and stopband characteristics.
14. Analyze the effect of cutoff frequency on filter performance.
15. Interpret the results and draw conclusions.

Matlab Program

```

clc;
clear;
close all;
Fs = 10000;      % Sampling Frequency
N = 4;           % Filter Order
Fc = [1000 2000 3000]; % Cutoff Frequencies

for k = 1:length(Fc)
    Wn = Fc(k)/(Fs/2);
    [b,a] = butter(N,Wn,'low');
    [H,f] = freqz(b,a,1024,Fs);

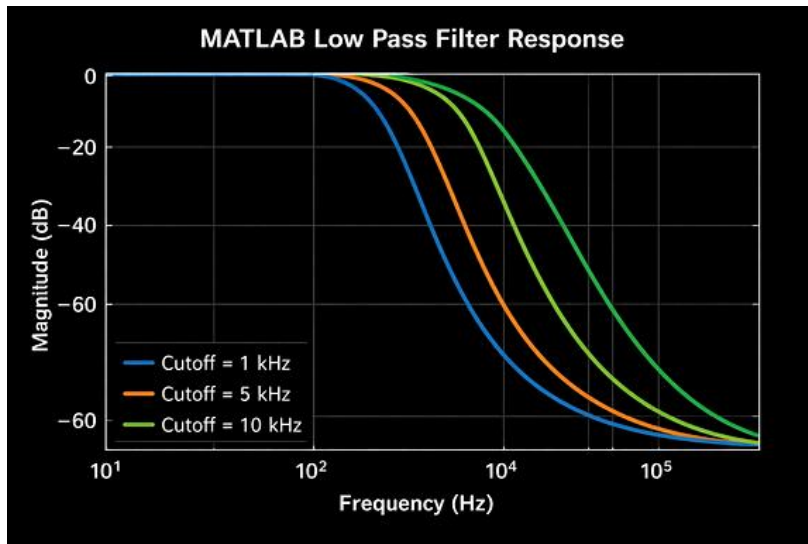
```

```

figure;
plot(f,20*log10(abs(H)),'LineWidth',2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title(['Magnitude Response of LPF (Fc = ',num2str(Fc(k)),' Hz)']);
end

```

Model Graph



Tabular Column

S.No	Sampling Frequency F_s (kHz)	Cutoff Frequency F_c (kHz)	Filter Order	Passband Response

Result

The Low Pass Filter (LPF) was successfully designed and simulated using MATLAB. The frequency response characteristics were analyzed for different cutoff frequencies. It was observed that frequencies below the cutoff frequency were passed with minimal attenuation, while frequencies above the cutoff frequency were significantly attenuated. Increasing the cutoff frequency increased the passband range, allowing a larger portion of the input signal spectrum to pass through the filter while suppressing unwanted high-frequency components.

Objective 2: To analyze the frequency response characteristics of the Low-Pass Filter using MATLAB by studying its gain, attenuation, and cutoff frequency behavior over a specified frequency range.

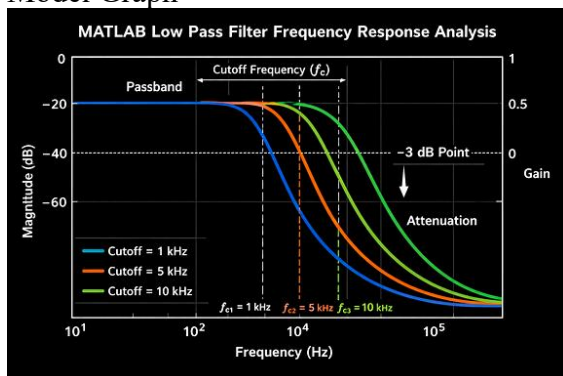
Procedure

1. Open MATLAB and create a new script file.
2. Clear the command window, workspace, and existing figures.
3. Define the sampling frequency, cutoff frequency, and filter order.
4. Normalize the cutoff frequency with respect to the Nyquist frequency.
5. Design the Butterworth Low-Pass Filter.
6. Obtain the filter coefficients.
7. Compute the frequency response using the `freqz()` function.
8. Calculate the magnitude response in dB.
9. Determine the attenuation characteristics.
10. Plot the magnitude response versus frequency.
11. Plot the attenuation versus frequency.
12. Repeat the analysis for different cutoff frequencies.
13. Compare the gain and attenuation characteristics.
14. Analyze the cutoff frequency behavior.
15. Interpret the filter performance and draw conclusions.

Matlab Program

```
clc;
clear;
close all;
Fs = 10000;          % Sampling Frequency
N = 4;               % Filter Order
Fc = [1000 2000 3000]; % Cutoff Frequencies
for k = 1:length(Fc)
    Wn = Fc(k)/(Fs/2);
    [b,a] = butter(N,Wn,'low');
    [H,f] = freqz(b,a,1024,Fs);
    Gain_dB = 20*log10(abs(H));
    figure;
    plot(f,Gain_dB,'LineWidth',2);
    grid on;
    xlabel('Frequency (Hz)');
    ylabel('Gain (dB)');
    title(['Frequency Response of LPF (Fc = ',num2str(Fc(k)),' Hz)']);
end
```

Model Graph



Tabular Column

Case	Sampling Frequency F_s (kHz)	Cutoff Frequency F_c (kHz)	Filter Order	Gain at Passband (dB)	Attenuation Beyond F_c

Result

The frequency response characteristics of the Low-Pass Filter were successfully analyzed using MATLAB. The filter exhibited nearly constant gain in the passband and significant attenuation in the stopband. It was observed that increasing the cutoff frequency widened the passband range, allowing more low-frequency components to pass through. The simulation verified the gain, attenuation, and cutoff frequency behavior of the Low-Pass Filter over the specified frequency range.

Question 1

A Low Pass Filter (LPF) is used in communication and microwave systems to pass signals below a specified cutoff frequency and attenuate signals above it. Simulate the LPF using MATLAB and analyze its frequency response characteristics under the following operating conditions.

Case 1

- Sampling Frequency (F_s) = 10 kHz
- Cutoff Frequency (F_c) = 1 kHz
- Filter Order (N) = 4

Case 2

- Sampling Frequency (F_s) = 10 kHz
- Cutoff Frequency (F_c) = 2 kHz
- Filter Order (N) = 4

Case 3

- Sampling Frequency (F_s) = 10 kHz
- Cutoff Frequency (F_c) = 3 kHz
- Filter Order (N) = 4

Tasks

1. Develop a MATLAB model of the Low Pass Filter (LPF) using the Butterworth filter design technique.
2. Design the LPF for each cutoff frequency and obtain the filter coefficients.
3. Simulate the frequency response and determine the passband and stopband characteristics.
4. Plot suitable graphs representing:
 - Magnitude Response (dB) vs Frequency
 - Phase Response vs Frequency
5. Compare the frequency response characteristics for all three cutoff frequencies and interpret the effect of cutoff frequency on filter performance.

Question 2

A Butterworth Low-Pass Filter (LPF) is used to allow low-frequency signals to pass while attenuating high-frequency signals. Simulate the frequency response of the LPF using MATLAB and analyze its gain and attenuation characteristics under the following operating conditions.

Case 1

- Sampling Frequency (F_s) = 10 kHz
- Cutoff Frequency (F_c) = 1 kHz
- Filter Order (N) = 4

Case 2

- Sampling Frequency (F_s) = 10 kHz
- Cutoff Frequency (F_c) = 2 kHz
- Filter Order (N) = 4

Case 3

- Sampling Frequency (F_s) = 10 kHz
- Cutoff Frequency (F_c) = 3 kHz
- Filter Order (N) = 4

Tasks

1. Develop a MATLAB model of the Low-Pass Filter using the Butterworth filter design technique.
2. Simulate the frequency response of the filter over the specified frequency range.
3. Determine the gain and attenuation characteristics at different frequencies.
4. Plot suitable graphs representing:
 - Magnitude Response (Gain in dB) vs Frequency
 - Attenuation vs Frequency
5. Compare the frequency response characteristics for all three cutoff frequencies and interpret the effect of cutoff frequency on passband, transition band, and stopband performance.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 7.2	Design of HPF for different frequency using Matlab
DATE:	

Objectives

i) To design and simulate a High Pass Filter (HPF) using MATLAB for different cutoff frequencies and study its frequency response characteristics. This helps in understanding the filtering of unwanted low-frequency components while allowing high-frequency signals to pass.

ii) To analyze the effect of varying cutoff frequencies on attenuation, insertion loss, and transmission characteristics of the High Pass Filter. The study enables the evaluation of filter performance under different operating conditions.

Objective 1: To design a High-Pass Filter (HPF) using MATLAB by selecting and implementing different cutoff frequencies suitable for various signal conditions.

Procedure

1. Open MATLAB and create a new script file.
2. Clear the command window, workspace, and close all figures.
3. Define the sampling frequency (F_s), cutoff frequency (F_c), and filter order (N).
4. Normalize the cutoff frequency with respect to the Nyquist frequency.
5. Design the High-Pass Filter using the Butterworth filter design method.
6. Obtain the filter coefficients using the `butter()` function.
7. Compute the frequency response using the `freqz()` function.
8. Calculate the magnitude response in dB.
9. Determine the phase response of the filter.
10. Plot the magnitude response versus frequency.
11. Plot the phase response versus frequency.
12. Repeat the procedure for all cutoff frequencies.
13. Compare the passband and stopband characteristics.
14. Analyze the effect of cutoff frequency on filter behavior.
15. Interpret the results and draw conclusions.

Matlab Program:

```
clc;
clear;
```

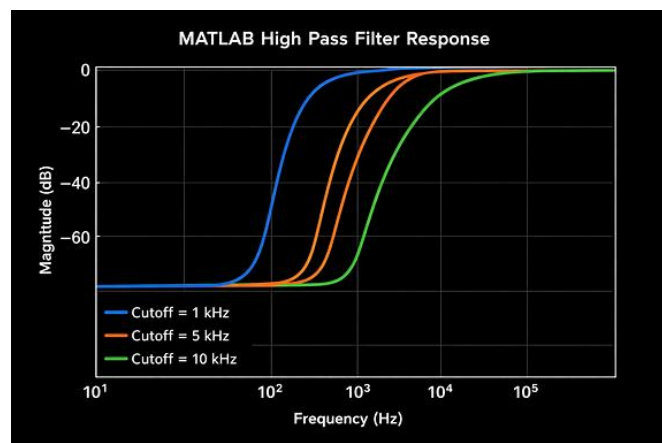
```

close all;

Fs = 10000;          % Sampling Frequency
N = 4;              % Filter Order
Fc = [1000 2000 3000]; % Cutoff Frequencies
for k = 1:length(Fc)
    Wn = Fc(k)/(Fs/2);
    % High Pass Filter Design
    [b,a] = butter(N,Wn,'high');
    % Frequency Response
    [H,f] = freqz(b,a,1024,Fs);
    % Magnitude Response
    figure;
    plot(f,20*log10(abs(H)),'LineWidth',2);
    grid on;
    xlabel('Frequency (Hz)');
    ylabel('Magnitude (dB)');
    title(['HPF Magnitude Response (Fc = ',num2str(Fc(k)),' Hz)']);
end

```

Model Graph



Tabular Column

S.No	Sampling Frequency (kHz)	Fs	Cutoff Frequency Fc (kHz)	Filter Order	Frequencies Passed

Result

The High-Pass Filter (HPF) was successfully designed and simulated using MATLAB. The frequency response analysis showed that frequencies below the cutoff frequency were significantly attenuated, while frequencies above the cutoff frequency were allowed to pass with minimal attenuation. It was observed that increasing the cutoff frequency increased the stopband range and reduced the range of frequencies passed by the filter. Thus, the HPF effectively removed unwanted low-frequency components and retained the desired high-frequency signals.

Objective 2: To analyze the frequency response characteristics of the High-Pass Filter (HPF) by plotting and studying its magnitude and phase response using MATLAB.

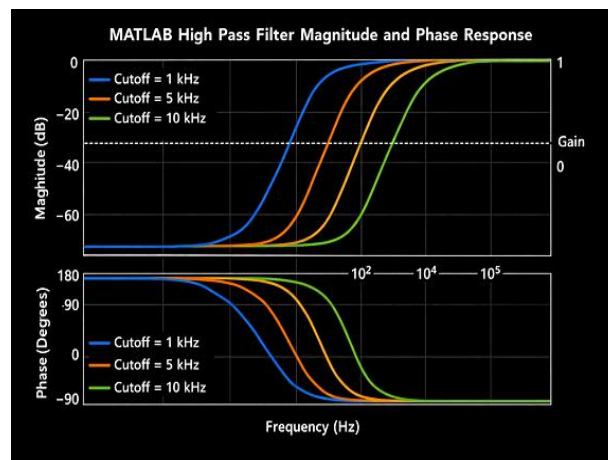
Procedure

1. Open MATLAB and create a new script file.
2. Clear the command window, workspace, and close all figures.
3. Define the sampling frequency, cutoff frequency, and filter order.
4. Normalize the cutoff frequency with respect to the Nyquist frequency.
5. Design the Butterworth High-Pass Filter using the `butter()` function.
6. Obtain the filter coefficients.
7. Compute the frequency response using the `freqz()` function.
8. Calculate the magnitude response in dB.
9. Calculate the phase response in degrees.
10. Plot the magnitude response versus frequency.
11. Plot the phase response versus frequency.
12. Repeat the simulation for all cutoff frequencies.
13. Compare the passband and stopband characteristics.
14. Analyze the effect of cutoff frequency on the frequency response.
15. Interpret the results and draw conclusions.

Matlab Program

```
clc;
clear;
close all;
Fs = 10000;          % Sampling Frequency
N = 4;              % Filter Order
Fc = [1000 2000 3000]; % Cutoff Frequencies
for k = 1:length(Fc)
    Wn = Fc(k)/(Fs/2);
    % High Pass Filter Design
    [b,a] = butter(N,Wn,'high');
    % Frequency Response
    [H,f] = freqz(b,a,1024,Fs);
    % Magnitude Response
    figure;
    plot(f,20*log10(abs(H)),'LineWidth',2);
    grid on;
    xlabel('Frequency (Hz)');
    ylabel('Magnitude (dB)');
    title(['Magnitude Response of HPF (Fc = ',num2str(Fc(k)),' Hz)']);
    % Phase Response
    figure;
    plot(f,angle(H)*180/pi,'LineWidth',2);
    grid on;
    xlabel('Frequency (Hz)');
    ylabel('Phase (Degrees)');
    title(['Phase Response of HPF (Fc = ',num2str(Fc(k)),' Hz)']);
end
```

Model Graph



Tabular Column

S.No	Sampling Frequency (kHz)	Fs	Cutoff Frequency (kHz)	Fc	Filter Order	Magnitude Response

Result

The frequency response characteristics of the High-Pass Filter (HPF) were successfully analyzed using MATLAB. The magnitude response showed significant attenuation of frequencies below the cutoff frequency and minimal attenuation of frequencies above the cutoff frequency. The phase response varied with frequency and filter order. It was observed that increasing the cutoff frequency increased the stopband range and shifted the passband toward higher frequencies. The simulation verified the gain, attenuation, and phase characteristics of the High-Pass Filter over the specified frequency range.

Question 1

A High-Pass Filter (HPF) is used to attenuate low-frequency components while allowing high-frequency signals to pass. Design and simulate the HPF using MATLAB and analyze its frequency response characteristics under the following operating conditions.

Case 1

- Sampling Frequency (F_s) = 10 kHz
- Cutoff Frequency (F_c) = 1 kHz
- Filter Order (N) = 4

Case 2

- Sampling Frequency (F_s) = 10 kHz
- Cutoff Frequency (F_c) = 2 kHz
- Filter Order (N) = 4

Case 3

- Sampling Frequency (F_s) = 10 kHz
- Cutoff Frequency (F_c) = 3 kHz
- Filter Order (N) = 4

Tasks

1. Develop a MATLAB model of the High-Pass Filter using the Butterworth filter design technique.
2. Design the HPF for each cutoff frequency and obtain the filter coefficients.
3. Simulate the frequency response and determine the passband and stopband characteristics.
4. Plot suitable graphs representing:
 - Magnitude Response (dB) vs Frequency
 - Phase Response vs Frequency
5. Compare the frequency response characteristics for all three cutoff frequencies and interpret

the effect of cutoff frequency on filter performance.

Question 2

A High-Pass Filter (HPF) is used to attenuate low-frequency components and allow high-frequency components to pass through. Simulate the HPF using MATLAB and analyze its magnitude and phase response characteristics under the following operating conditions.

Case 1

- Sampling Frequency (F_s) = 10 kHz
- Cutoff Frequency (F_c) = 1 kHz
- Filter Order (N) = 4

Case 2

- Sampling Frequency (F_s) = 10 kHz
- Cutoff Frequency (F_c) = 2 kHz
- Filter Order (N) = 4

Case 3

- Sampling Frequency (F_s) = 10 kHz
- Cutoff Frequency (F_c) = 3 kHz
- Filter Order (N) = 4

Tasks

1. Develop a MATLAB model of the High-Pass Filter using the Butterworth filter design technique.
2. Simulate the frequency response of the HPF over the specified frequency range.
3. Determine the magnitude response and phase response characteristics.
4. Plot suitable graphs representing:
 - Magnitude Response (dB) vs Frequency
 - Phase Response vs Frequency
5. Compare the frequency response characteristics for all three cutoff frequencies and interpret the effect of cutoff frequency on gain, attenuation, and phase behavior.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 8.1	Frequency and Wavelength measurement using Reflex Klystron
DATE:	

Objectives

i) To determine the frequency and wavelength of microwave signals generated by a Reflex Klystron using a microwave test bench setup. This helps in understanding the basic operating principle of microwave signal generation.

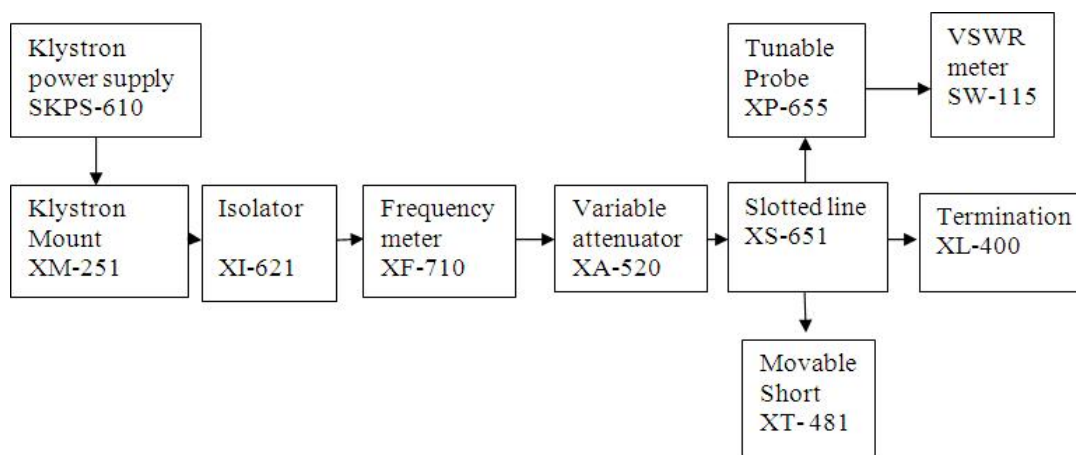
ii) To measure the resonant modes and identify the operating frequency of the Reflex Klystron by analyzing the standing wave pattern using appropriate measurement instruments. This enables accurate characterization of microwave output using matlab simulation

i) To determine the frequency and wavelength of microwave signals generated by a Reflex Klystron using a microwave test bench setup. This helps in understanding the basic operating principle of microwave signal generation.

APPARATUS REQUIRED:

1. Reflex klystron power supply
2. klystron Mount
3. Isolator
4. Frequency Meter
5. Variable Attenuator
6. Slotted section
7. Wave Guide Stand
8. VSWR Meter
9. Digital Storage Oscilloscope (DSO)
10. Matched termination
11. Movable short
12. BNC Cable

Block Diagram



THEORY

In a rectangular waveguide operating in the dominant TE_{10} mode, the free-space wavelength (λ_0), the guide wavelength (λ_g), and the cutoff wavelength (λ_c) are related as:

$$\frac{1}{\lambda_0^2} = \frac{1}{\lambda_g^2} + \frac{1}{\lambda_c^2}$$

In a rectangular waveguide operating in the dominant TE_{10} mode, the free-space wavelength (λ_0), the guide wavelength (λ_g), and the cutoff wavelength (λ_c) are related as:

$$\frac{1}{\lambda_g^2} = \frac{1}{\lambda_0^2} - \frac{1}{\lambda_c^2}$$

Free-space wavelength (λ_0):

The wavelength of the electromagnetic wave in free space.

$$\lambda_0 = \frac{c}{f}$$

where c is the speed of light (3×10^8 m/s) and f is the frequency of the signal.

Guide wavelength (λ_g):

- The wavelength of the electromagnetic wave inside the waveguide. It is longer than the free-space wavelength due to the waveguide's cutoff effect.

Cutoff wavelength (λ_c):

- The longest wavelength (or lowest frequency) that can propagate in the waveguide. For the dominant TE_{10} mode, the cutoff wavelength is:

$$\lambda_c = 2a$$

where a is the broad dimension of the waveguide.

PROCEDURE

1. Set up the components and equipment as shown in fig. above.
2. Set the variable Attenuator at the minimum attenuation position.
3. Keep the control knobs of VSWR Meter as bellow: Range - 50db
 - Input switch - crystal low impedance
 - Meter switch - Normal position
 - Gain - Mid position
 - Keep the control knobs of klystron power supply as bellow
 - Beam Voltage - OFF
 - Mod switch - AM
 - Beam voltage knob - fully anticlockwise Reflector Voltage - fully clockwise
 - AM amplitude knob - Around fully clockwise AM Frequency knob - Around mid
4. Switch ON the klystron power supply, VSWR meter and cooling fan.

- Switch ON the beam voltage switch and rotate the beam voltage knob clockwise up to 200v deflection in meter.
- 5. Keep the AM-MOD amplitude knob and AM-FREQ knob at the mid –position.
- 6. Rotate the reflector voltage knob to get deflection in VSWR meter.
- 7. Rotate the AM-MOD amplitude knob to get the maximum output in VSWR meter
- 8. Maximize the deflection with frequency knob to get the maximum output in VSWR meter
- 9. Tune the probe for maximum deflection in VSWR meter
- 10. Tune the frequency meter knob to get a „dip“ on the VSWR meter scale and note down the frequency directly from the frequency meter
- 11. Replace the termination with movable short and detune the frequency meter
- 12. By moving the probe along the slotted line vary the deflection on VSWR meter.
- 13. Move the probe to a minimum deflection position. Note and record the probe position
- 14. Move the probe to next minimum position and record the probe position again
- 15. Calculate guide wavelength as twice the distance between two successive minimum positions obtained as above.
- 16. Measure the wave guide inner broad dimension „a“ which will be around 22.86 mm for X-band

PRECAUTIONS

- Beam voltage should not exceed 250V
- Reflected voltage should not exceed 270V
- Fan should be operated to reduce the heat for reflex klystron
- Every 20 minutes switch of the supply and set the initial conditions.
- Carefully follow the specifications of the devices and equipment.

Result Tabulation

S.No	Frequency (GHz)	Wavelength (λ) [cm]	Remarks
1	8	3.75	Initial frequency setting
2	9	3.16	Adjusted frequency
3	10	3	Near maximum frequency

Calculation of Wavelength:

For each frequency reading, the corresponding wavelength can be calculated using the

$$\lambda = \frac{c}{f}$$

formula

Where:

- $C=3 \times 10^8$ m/s
- f is in Hz (for GHz, multiply by 10^9)

For example:

- For a frequency of 8.00 GHz:

$$\lambda = \frac{3 \times 10^8}{8.00 \times 10^9} = 3.75 \text{ cm}$$

Example for Understanding

Given

- Distance between two successive minima = 2.25 cm

Since:

$$\lambda_g = 2 \times 2.25 = 4.5 \text{ cm}$$

For X-band waveguide WR-90:

$$a = 2.286 \text{ cm}$$

$$\lambda_c = 2a = 4.572 \text{ cm}$$

Substituting:

$$\lambda = \frac{4.5 \times 4.572}{\sqrt{4.5^2 + 4.572^2}}$$

$$\lambda = 3.21 \text{ cm}$$

$$f = \frac{3 \times 10^8}{3.21 \times 10^{-2}}$$

$$f = 9.35 \text{ GHz}$$

Result

- Guide wavelength = 4.5 cm
- Free-space wavelength = 3.21 cm
- Frequency = 9.35 GHz

Sample Observation Table

Minima Position (cm)	Next Minima Position (cm)	$\lambda_g/2$ (cm)	λ_g (cm)
3.2	5.45	2.25	4.50
5.45	7.70	2.25	4.50
7.70	9.95	2.25	4.50

Average Guide Wavelength:

$$\lambda_g = 4.5 \text{ cm}$$

ii) To measure the resonant modes and identify the operating frequency of the Reflex Klystron by analyzing the standing wave pattern using MATLAB simulation. This enables accurate characterization of microwave output characteristics.

Procedure

1. Initialize the MATLAB environment and define the operating frequency range of the Reflex Klystron.
2. Simulate the microwave output power variation with frequency around the nominal operating frequency.
3. Generate resonance peaks corresponding to different resonant modes of the Reflex

Klystron.

4. Calculate the output power for each frequency point.
5. Plot Output Power versus Frequency to obtain the resonance curve.
6. Identify the frequency corresponding to the maximum output power.
7. Determine the resonant mode and operating frequency from the resonance peak.
8. Analyze the standing wave pattern and resonance characteristics.
9. Tabulate the resonance frequencies and corresponding output powers.
10. Verify the operating mode from the simulated resonance graph.

MATLAB Code

```
clc;
clear;
close all;

% Frequency Range (GHz)
f = 8:0.01:10;

% Simulated Resonance Curve
P = exp(-((f-9).^2)/0.02);

% Find Resonance Frequency
[maxPower,index] = max(P);
resonantFrequency = f(index);

fprintf('Resonant Frequency = %.2f GHz\n',resonantFrequency);

% Plot
figure;
plot(f,P,'LineWidth',2);
grid on;

xlabel('Frequency (GHz)');
ylabel('Normalized Output Power');
title('Resonance Characteristics of Reflex Klystron');

hold on;
plot(resonantFrequency,maxPower,'ro','MarkerSize',10,...
    'MarkerFaceColor','r');

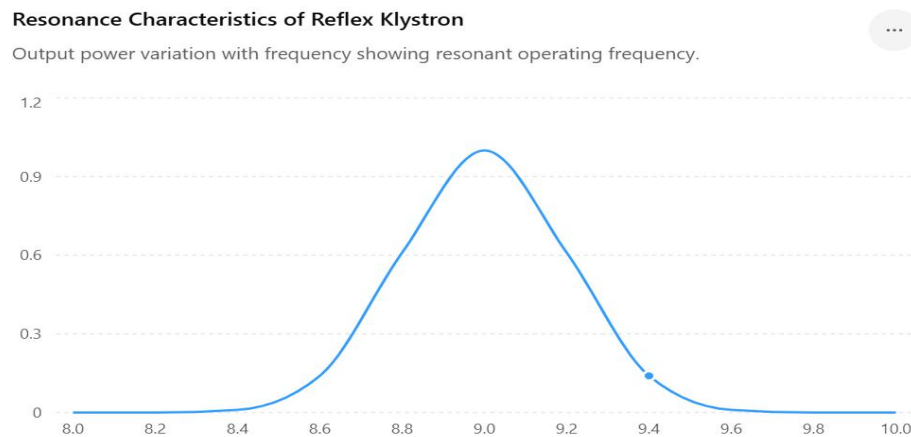
legend('Resonance Curve','Operating Frequency');
```

Tabulation:

Frequency (GHz)	Output Power

Model Graph

The resonance curve obtained from the MATLAB simulation will appear as follows:



Result

The resonance characteristics of the Reflex Klystron were successfully simulated using MATLAB. The maximum output power occurs at 9 GHz, indicating the resonant operating frequency of the Reflex Klystron. The resonance curve confirms the presence of a resonant mode at this frequency and demonstrates the microwave output characteristics of the device.

Question 1

A Reflex Klystron operates based on the interaction between the electron beam and the resonant cavity. The oscillation mode depends on the electron transit time between the cavity gap and the repeller electrode.

Case 1

Operating Frequency = 8 GHz

Case 2

Operating Frequency = 9 GHz

Case 3

Operating Frequency = 10 GHz

Tasks

1. Calculate the RF period corresponding to 10 GHz.
2. Determine the transit time for mode numbers $n=0$ to $n=888$.
3. Tabulate the mode number and corresponding transit time values.

4. Obtain frequency by observing minima values.
5. Analyze the effect of operating frequency on oscillation mode behavior.

Question 2

To measure the resonant modes and identify the operating frequency of a Reflex Klystron by analyzing the standing wave pattern and resonance characteristics using MATLAB simulation.

The output power of a Reflex Klystron varies with frequency and exhibits resonance peaks at specific operating frequencies. The resonant frequency corresponds to the maximum output power and indicates the operating mode of the device.

Case 1

Frequency Range = 7 GHz to 9 GHz

Resonance Frequency = 8 GHz

Case 2

Frequency Range = 8 GHz to 10 GHz

Resonance Frequency = 9 GHz

Case 3

Frequency Range = 9 GHz to 11 GHz

Resonance Frequency = 10 GHz

Tasks

1. Simulate the output power variation over the given frequency range.
2. Plot Frequency versus Normalized Output Power.
3. Determine the frequency corresponding to the maximum output power.
4. Tabulate the frequency and output power values.
5. Identify the resonant mode from the resonance curve.
1. Compare the resonance characteristics obtained in Cases 1 and 2 and comment on the shift in operating frequency.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 8.2	Voltage Standing Wave Ratio (VSWR) measurement
DATE:	

Objectives

i) To measure the Voltage Standing Wave Ratio (VSWR) in a microwave transmission line using a slotted line setup and detector probe. This helps in understanding impedance matching conditions in the transmission system.

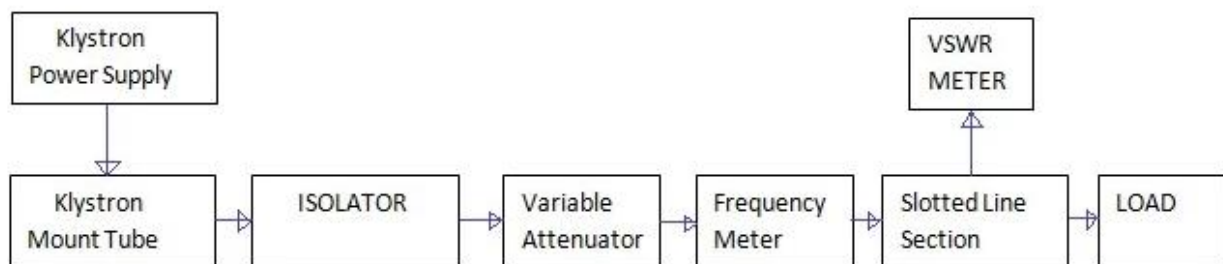
ii) To determine the reflected and incident wave components by analyzing the standing wave pattern along the transmission line. This enables evaluation of mismatch between load and transmission line using matlab.

I) To measure the Voltage Standing Wave Ratio (VSWR) in a microwave transmission line using a slotted line setup and detector probe. This helps in understanding impedance matching conditions in the transmission system.

APPARATUS REQUIRED:

13. Reflex klystron power supply
14. klystron Mount
15. Isolator
16. Frequency Meter
17. Variable Attenuator
18. Slotted section
19. Wave Guide Stand
20. VSWR Meter
21. Digital Storage Oscilloscope (DSO)
22. Matched termination
23. Movable short
24. BNC Cable

Blockdiagram:



THEORY:

It is a ratio of maximum voltage to minimum voltage along a transmission line is called VSWR, as ratio of maximum to minimum current. SWR is measure of mismatch between load and line. The electromagnetic field at any point of transmission line may be considered as the sum of two traveling waves: the 'Incident Wave' propagates from generator and the

reflected wave propagates towards the generator. The reflected wave is set up by reflection of incident wave from a discontinuity on the line or from the load impedance. The magnitude and phase of reflected wave depends upon amplitude and phase of the reflecting impedance. The superposition of two traveling waves, gives rise to standing wave along with the line. The maximum field strength is found where two waves are in phase and minimum where the line adds in opposite phase. The distance between two successive minimum (or maximum) is half the guide wavelength on the line. The ratio of electrical field strength of reflected and incident wave is called reflection between maximum and minimum field strength along the line.

Hence VSWR denoted by S is

$$S = \frac{E_{max}}{E_{min}}$$

$$= \frac{|E_I| + |E_R|}{|E_I| - |E_R|}$$

Where, E_I = incident voltage

E_R = reflected voltage

Reflection Coefficient, ρ is

$$\rho = \frac{E_R}{E_I} = \frac{Z - Z_0}{Z + Z_0}$$

Where, Z is the impedance at a point on line,

Z_0 is characteristic Impedance.

The above equation gives following equation $|\rho| = -1 \text{ to } +1$

Procedure:

1. Set the equipments as figure –(5)
2. Keep variable attenuator in the minimum attenuation position
3. Keep the control knob of VSWR meter as below

Range db	-	40db to 50db
Input Switch	-	Low impedance
Meter Switch	-	Normal
Gain	-	Mid position
4. Keep the control knob of Klystron power supply

Beam voltage	-	Off
Mod-switch	-	AM
Beam voltage knob	-	Full anti clockwise
Reflector voltage knob-		Full clockwise

Am-amplitude knob - Full clockwise

Am frequency & amplitude knob - Mid position

5. Switch on the klystron power supply, VSWR meter & cooling fan.
6. Switch on the beam voltage switch and set beam voltage at 260V.
7. Rotate the reflector voltage knob to get deflection in VSWR meter.
8. Tune the output by tuning the reflector voltage, amplitude and frequency of am modulation.
9. Tune the plunger of klystron mount and probe for maximum deflection in VSWR meter.
10. If required change the range db switch variable attenuator position and gain control knob to get deflection in the scale of VSWR meter.
11. As we move probe along the slotted line, the deflection will change.
12. Measurement of low and medium VSWR.

Measurement of low and medium V.S.W.R.

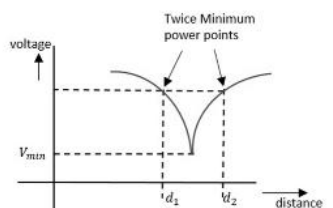
- Move the probe along the slotted line to get maximum deflection in VSWR meter.
- Adjust the VSWR meter gain control knob or variable attenuator until the meter indicated 1.0 on normal VSWR scale.
- Keep all the control knobs as it is, move the probe to next minimum position. Read the VSWR on scale.
- Repeat the above step for change of SS tuner probe depth and record the corresponding SWR.
- The VSWR is between 3.2 and 10, change the range db to next higher position and read the VSWR on second VSWR scale of 3 to 10.

Measurement of high VSWR

Measurement of high VSWR: Select “Unmatched Load” to terminate the slotted line by pressing the button.

- Use slider to fix the value of “Resistance” and “Reactance” of the load.
- Locate the position of V_{min} and take it as a reference.(If VSWR meter is used in actual experiment, set the output so that meter reads 3dB).
- Move the slider (probe of slotted line) along the slotted line on either side of V_{min} so that the reading is 3 db below the reference i.e. 0 db. Record the probe positions and obtain the distance between the two. Determine the VSWR using equation (2).
- The simulated value for VSWR can be seen by clicking the buttons “Technique used to calculate VSWR 1 & 2”.
- Then match the calculated value with the value displayed in the simulated VSWR

Graph:



Observation: Single Minima

Frequency (f)	Power Maxima (P _{max}) at	Power Minima (P _{min}) at	VSWR
2.0GHz	1.6W	0.4W	2
3.0GHz	2.0W	0.8W	1.58

Double Minima

Frequency (f)	Position of Minima (x1, x2)	Power Maxima (P _{max}) at	Power Minima (P _{min}) at	VSWR
2.0GHz	10.0cm, 12.5cm	1.5W	0.3W	2.24

Calculation for Example

1. Measure successive minima:
 $x_2 - x_1 = 12.5 - 10.0 = 2.5$ cm (Half-wavelength is found to be 2.5 cm)
2. Power ratio:

$$\sqrt{\frac{P_{\max}}{P_{\min}}} = \sqrt{\frac{1.5}{0.3}} = \sqrt{5} \approx 2.24$$

Example for Understanding (Simple Numerical Case)

Problem

In a Reflex Klystron experiment, the distance between two successive minima is 2.0 cm. Waveguide cutoff wavelength is 4.5 cm.

Find:

- Guide wavelength (λ_g)
- Free-space wavelength (λ)
- Frequency (f)

Step 1: Guide Wavelength

Distance between minima = $\lambda_g / 2$

$$\lambda_g = 2 \times 2.0 = 4.0 \text{ cm}$$

Step 2: Free-space wavelength

$$\lambda = \frac{\lambda_g \lambda_c}{\sqrt{\lambda_g^2 + \lambda_c^2}}$$

$$\lambda = \frac{4.0 \times 4.5}{\sqrt{4.0^2 + 4.5^2}}$$

$$\lambda = \frac{18}{\sqrt{16 + 20.25}} = \frac{18}{6.02}$$

$$\lambda = 2.99 \text{ cm}$$

Step 3: Frequency

$$f = \frac{3 \times 10^8}{2.99 \times 10^{-2}}$$
$$f = 10.03 \text{ GHz}$$

Objective ii) To determine the reflected and incident wave components by analyzing the standing wave pattern along the transmission line.

Procedure

1. Initialize MATLAB and define the transmission line length.
2. Specify the incident wave amplitude.
3. Define the reflection coefficient representing the load mismatch.
4. Generate the standing wave pattern along the transmission line.
5. Calculate the voltage distribution resulting from incident and reflected waves.
6. Plot Voltage Amplitude versus Distance.
7. Identify voltage maxima and minima from the standing wave pattern.
8. Determine the reflected wave component.
9. Calculate SWR and evaluate the degree of mismatch.
10. Analyze the effect of reflection coefficient on standing wave formation.

MATLAB Code

```
clc;
clear;
close all;

% Transmission Line Length
z = linspace(0,1,1000);

% Incident Wave Amplitude
Vi = 1;

% Reflection Coefficient
Gamma = 0.4;
% Phase Constant
beta = 2*pi*5;

% Standing Wave Pattern
V = abs(Vi*(1 + Gamma*exp(-1j*2*beta*z)));

% Maximum and Minimum Voltage
Vmax = max(V);
Vmin = min(V);

% SWR Calculation
```

```

SWR = Vmax/Vmin;

fprintf('Maximum Voltage = %.3f\n',Vmax);
fprintf('Minimum Voltage = %.3f\n',Vmin);
fprintf('SWR = %.3f\n',SWR);

% Plot
figure;
plot(z,V,'LineWidth',2);
grid on;

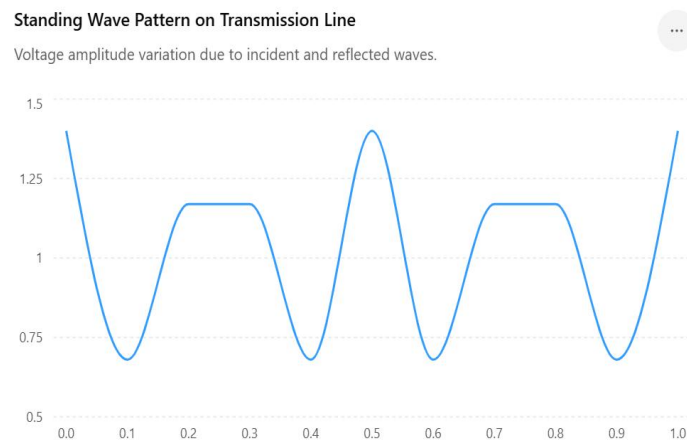
xlabel('Distance along Transmission Line (m)');
ylabel('Voltage Amplitude');
title('Standing Wave Pattern');

```

Distance (m)	Voltage Amplitude

Model Graph

The standing wave pattern obtained from MATLAB simulation will appear as:



Result

The standing wave pattern along the transmission line was successfully simulated using MATLAB. The incident and reflected wave components were identified from the voltage maxima and minima. The calculated SWR indicates the degree of mismatch between the transmission line and the load, thereby enabling characterization of transmission line performance.

Question 1

To determine the reflected and incident wave components by analyzing the standing wave pattern along a transmission line and evaluate the mismatch between the load and transmission line using MATLAB simulation.

A microwave transmission line carries an incident wave of unit amplitude. Due to impedance mismatch at the load, a portion of the wave is reflected, resulting in the formation of a standing wave pattern. The reflection coefficient (Γ) determines the magnitude of the reflected wave and the Standing Wave Ratio (SWR).

Case 1

Reflection Coefficient (Γ) = 0.2

Case 2

Reflection Coefficient (Γ) = 0.4

Case 3

Reflection Coefficient (Γ) = 0.6

Tasks

1. Simulate the standing wave pattern for a highly mismatched load.
2. Determine the reflected wave magnitude and voltage distribution.
3. Calculate $V_{\max}$, $V_{\min}$, and SWR.
4. Plot the standing wave pattern along the transmission line.
5. Compare the SWR values obtained in Cases 1 and 2.
6. Discuss the effect of increasing reflection coefficient on transmission line performance and power transfer efficiency.

Question 2

To analyze the effect of load impedance mismatch on reflected power and standing wave characteristics in a microwave transmission line using MATLAB simulation.

In a microwave transmission system, the mismatch between the load impedance and characteristic impedance causes reflections, resulting in standing waves along the transmission line. The degree of mismatch can be evaluated using the reflection coefficient, reflected power, and Standing Wave Ratio (SWR).

Case 1

Characteristic Impedance $Z_0=50\Omega$

Load Impedance $Z_L=75\Omega$

Case 2

Characteristic Impedance $Z_0=50\Omega$

Load Impedance $Z_L=100\Omega$

Case 3

Characteristic Impedance $Z_0=50\Omega$

Load Impedance $Z_L=150\Omega$

Tasks

1. Calculate the reflection coefficient (Γ).
2. Determine the reflected power percentage.
3. Calculate the SWR and identify voltage maxima and minima.
4. Simulate the standing wave pattern using MATLAB.
5. Plot Distance versus Voltage Amplitude.
6. Analyze the impact of severe impedance mismatch on microwave power transmission efficiency.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 9.1	Radiation Pattern Simulation of Dipole and Strip Line Antennas
DATE:	

Objectives

- i) To plot and analyze the radiation pattern of a Dipole Antenna using MATLAB by simulating its directional field distribution.
- ii) To Analyze the Beamwidth and Directional Characteristics of a Stripline Antenna
- I) To plot and analyze the radiation pattern of a Dipole Antenna using MATLAB by simulating its directional field distribution. This helps in understanding the antenna's radiation characteristics, including its gain, directivity, and spatial signal propagation behavior.

THEORETICAL BACKGROUND

A dipole antenna is one of the most widely used antennas in microwave and communication systems. A half-wave dipole has a length approximately equal to half of the operating wavelength.

The normalized electric field pattern of a dipole antenna is given by:

$$E(\theta) = \frac{\cos\left(\frac{\pi}{2} \cos\theta\right)}{\sin\theta}$$

where:

- $E(\theta)$ = Radiation field intensity
- θ = Observation angle

The radiation pattern has maximum radiation perpendicular to the antenna axis and minimum radiation along the antenna axis.

SIMULATION REQUIREMENTS

Software:

- MATLAB R2020a or later

Components Required:

- MATLAB Software

- Dipole Antenna Radiation Equation

SIMULATION PROCEDURE

1. Open MATLAB.
2. Define the angle range from 0° to 360° .
3. Compute the radiation intensity.
4. Normalize the radiation pattern.
5. Plot the polar radiation pattern.
6. Observe the directional characteristics.

% Radiation Pattern Simulation of Dipole Antenna

```
clc;
clear;
close all;
```

```
% Angular Range
theta = linspace(0,2*pi,1000);
```

```
% Radiation Pattern of Half-Wave Dipole
E = abs(sin(theta));
```

```
% Normalization
E_norm = E/max(E);
```

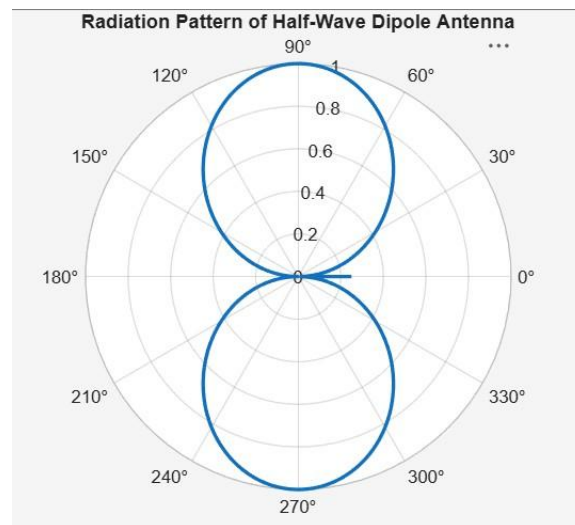
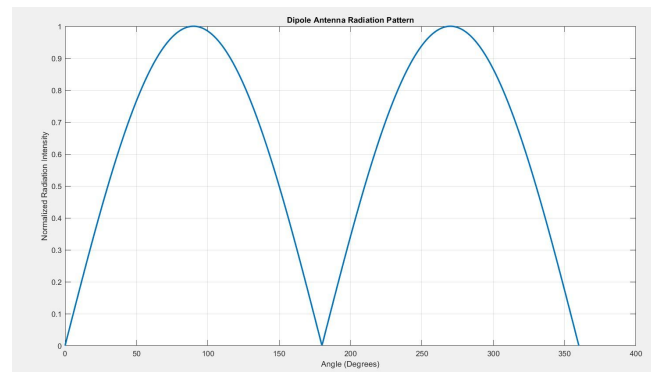
```
% Polar Plot
figure;
polarplot(theta,E_norm,'LineWidth',2);
title('Radiation Pattern of Half-Wave Dipole Antenna');
grid on;
```

```
% Cartesian Plot
figure;
plot(rad2deg(theta),E_norm,'LineWidth',2);
grid on;
xlabel('Angle (Degrees)');
ylabel('Normalized Radiation Intensity');
title('Dipole Antenna Radiation Pattern');
```

OBSERVATION TABLE

Parameter	Value
Antenna Type	Half-Wave Dipole
Operating Pattern	Bidirectional
Maximum Radiation	$\theta = 90^\circ$
Minimum Radiation	$\theta = 0^\circ, 180^\circ$

Model Graph



Radiation Pattern Generated Successfully

Maximum Radiation at Broadside Direction

Nulls Observed Along Antenna Axis

RESULT

The radiation pattern of the half-wave dipole antenna was successfully simulated using MATLAB. The antenna exhibited a bidirectional radiation pattern with maximum radiation perpendicular to the antenna axis.

Objective ii) To analyze the radiation characteristics, beamwidth, and directional properties of a stripline antenna.

Procedure

1. Open MATLAB software.
2. Define the angular range from 0° to 360°.

3. Model the stripline antenna radiation pattern.
4. Compute the radiation intensity at various angles.
5. Normalize the radiation pattern.
6. Plot the radiation pattern using MATLAB.
7. Observe the main lobe and side lobes.
8. Record the radiation characteristics.

MATLAB Program

% Radiation Pattern Simulation of Stripline Antenna

```
clc;  
clear;  
close all;
```

```
% Angular Range  
theta = linspace(0,2*pi,1000);
```

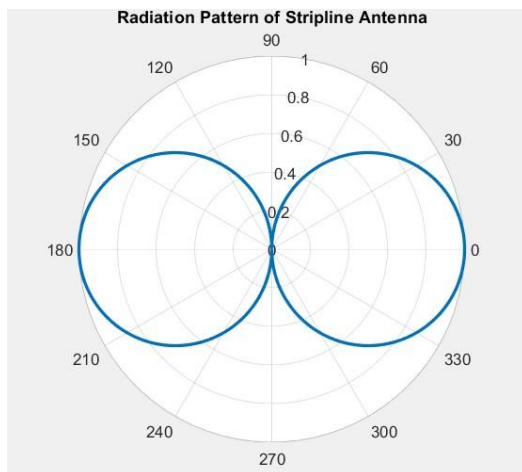
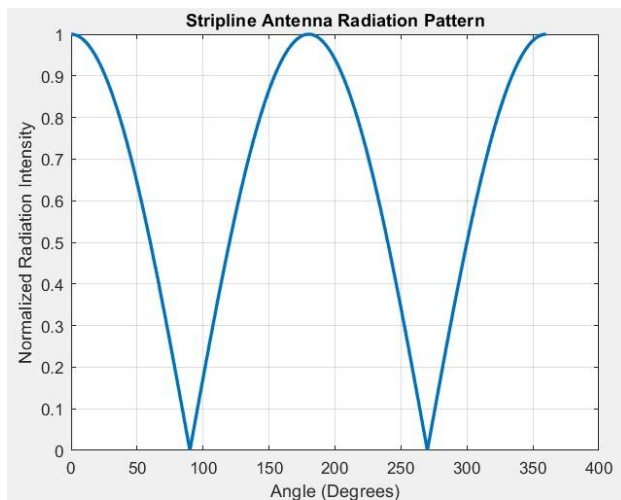
```
% Approximate Stripline Antenna Pattern  
E = abs(cos(theta));
```

```
% Normalization  
E_norm = E/max(E);
```

```
% Polar Plot  
figure;  
polarplot(theta,E_norm,'LineWidth',2);  
title('Radiation Pattern of Stripline Antenna');  
grid on;
```

```
% Cartesian Plot  
figure;  
plot(rad2deg(theta),E_norm,'LineWidth',2);  
grid on;  
xlabel('Angle (Degrees)');  
ylabel('Normalized Radiation Intensity');  
title('Stripline Antenna Radiation Pattern');
```

Model Graph



Tabular Column

Angle (°)	Radiation Intensity
0	
30	
60	
90	
120	
150	
180	

Example (Just for Understanding)

Example

At $\theta = 0^\circ$

$$\text{Radiation Intensity} = \cos(0^\circ) = 1$$

Normalized Radiation Intensity = 1

Result

The radiation pattern of the stripline antenna was successfully simulated using MATLAB. The obtained radiation plot showed the directional characteristics and radiation distribution of the stripline antenna.

Question 1

Analysis of the Radiation Pattern of a Dipole Antenna Using MATLAB

A half-wave dipole antenna is simulated using MATLAB to study its radiation characteristics and directional field distribution at different operating frequencies.

Case 1

- Operating Frequency = 300 MHz
- Antenna Length = 0.5λ
- Input Power = 10 W

Case 2

- Operating Frequency = 600 MHz
- Antenna Length = 0.5λ
- Input Power = 15 W

Case 3

- Operating Frequency = 900 MHz
- Antenna Length = 0.5λ
- Input Power = 20 W

Tasks

1. Develop a MATLAB model of the half-wave dipole antenna for the given operating frequencies.
2. Simulate the radiation pattern of the antenna for all three cases.
3. Plot the polar radiation pattern and analyze the field distribution.
4. Determine the gain and directivity of the dipole antenna.
5. Compare the radiation characteristics at different frequencies.
6. Identify the operating condition that provides maximum directional radiation.
7. Analyze the effect of operating frequency on antenna performance.

Question 2

Analysis of Directional Characteristics, Beam Shape, and Radiation Intensity Distribution of a Stripline Antenna Using MATLAB

A Stripline antenna is analyzed under different operating conditions to study its directional characteristics, beam shape, and radiation intensity distribution.

Case 1

- Antenna Length = 0.45λ
- Frequency = 500 MHz
- Maximum Radiation Intensity = 1.8 W/sr

Case 2

- Antenna Length = 0.50λ
- Frequency = 750 MHz
- Maximum Radiation Intensity = 2.4 W/sr

Case 3

- Antenna Length = 0.55λ
- Frequency = 1 GHz
- Maximum Radiation Intensity = 2.9 W/sr

Tasks

1. Simulate the dipole antenna using MATLAB for the given cases.
 2. Plot the radiation intensity pattern for each antenna configuration.
 3. Determine the beam shape and main lobe characteristics.
 4. Calculate the antenna gain and directivity.
 5. Compare the radiation intensity distribution for all three cases.
 6. Identify the antenna configuration that provides the best directional characteristics.
 7. Analyze the relationship between antenna length and radiation performance.
-
1. Determine the antenna suitable for high-gain microwave applications.
 2. Interpret the results and recommend the best antenna configuration for communication systems.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 9.2	Radiation Pattern Simulation of Microwave Horn Antenna using MATLAB
DATE:	

Objectives

- i) To simulate and analyze the radiation pattern of a Microwave E-plane Horn Antenna using MATLAB simulation.
- ii) To evaluate the performance of the H-plane Horn Antenna in terms of gain, directivity, beamwidth, and radiation efficiency.

1: To Simulate and analyze the radiation pattern of a Microwave E-Plane Horn Antenna using MATLAB simulation

Procedure

1. Open MATLAB software.
2. Define the angular range for radiation pattern analysis.
3. Model the horn antenna radiation pattern using an appropriate mathematical expression.
4. Compute the radiation intensity at different angles.
5. Normalize the radiation pattern.
6. Generate the polar radiation plot.
7. Observe the main lobe and side-lobe characteristics.
8. Record the radiation pattern observations.

MATLAB Program

% Radiation Pattern Simulation of Microwave E-plane Horn Antenna

```
clc;
clear;
close all;
```

```
% Angular Range
theta = linspace(-pi/2,pi/2,1000);
```

```
% Approximate Horn Antenna Radiation Pattern
E = (cos(theta)).^4;
```

```
% Normalize Pattern
E = abs(E);
E_norm = E/max(E);
```

```
% Polar Plot
figure;
polarplot(theta,E_norm,'LineWidth',2);
title('Radiation Pattern of Microwave Horn Antenna');
```

```
grid on;
```

```
% Cartesian Plot
```

```
figure;
```

```
plot(rad2deg(theta),E_norm,'LineWidth',2);
```

```
xlabel('Angle (Degrees)');
```

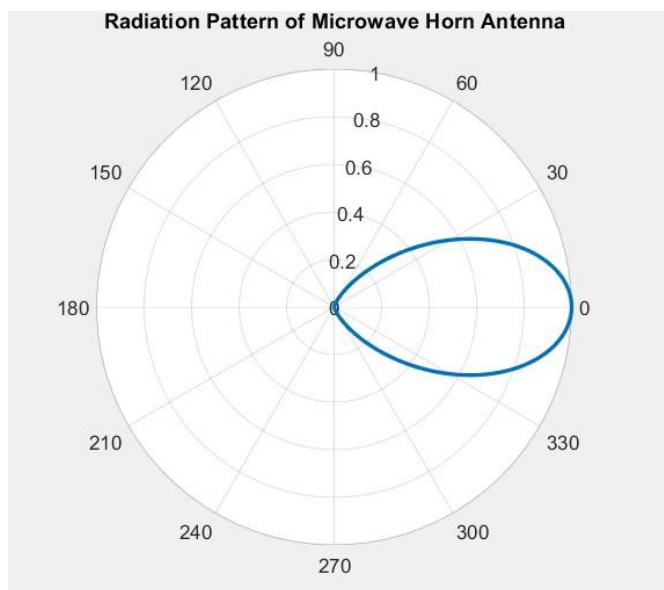
```
ylabel('Normalized Radiation Intensity');
```

```
title('Horn Antenna Radiation Pattern');
```

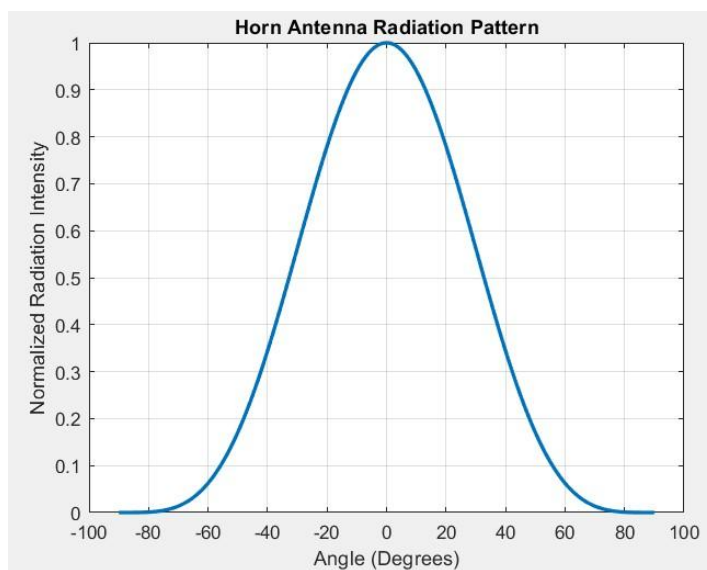
```
grid on;
```

Model Graph

- Polar radiation pattern of microwave horn antenna.



- Radiation intensity versus angle plot.



Tabular Column

Angle (°)	Radiation Intensity
-90	
-60	
-30	
0	
30	
60	

Example (Just for Understanding)

Example

At $\theta = 0^\circ$

Radiation Intensity = $\cos^4(0^\circ)$

= 1

Normalized Radiation Intensity = 1

Result

The radiation pattern of the microwave horn antenna was successfully simulated using MATLAB. The obtained radiation plot showed a highly directional main lobe with reduced side-lobe levels, demonstrating the directional characteristics of the horn antenna.

Objective 2: To Analyze the Directivity and Beamwidth of a Microwave Horn Antenna

Theory

An H-plane horn antenna is formed by flaring a rectangular waveguide in the magnetic field (H-field) plane. The radiation pattern is narrower in the H-plane due to the increased aperture dimension.

The normalized field pattern can be approximated for the sinc radiation pattern of the horn aperture.

Procedure

1. Open MATLAB software.
2. Define the horn antenna radiation pattern.
3. Generate the normalized radiation pattern.
4. Determine the maximum radiation intensity.
5. Identify the direction of maximum radiation.

6. Analyze the beamwidth of the main lobe.
7. Plot the radiation intensity distribution.
8. Record the directivity characteristics.

MATLAB Program

% Analysis of Microwave H-plane Horn Antenna Characteristics

```

Clc;
clear;
close all;

% Parameters
f = 10e9; % Frequency = 10 GHz
c = 3e8; % Speed of light
lambda = c/f; % Wavelength

a = 6*lambda; % Aperture width

% Angular range
theta = -90:0.1:90;
theta_rad = deg2rad(theta);

% Wave number
k = 2*pi/lambda;

% H-plane radiation pattern
u = (k*a/2).*sin(theta_rad);

E = ones(size(u));

for i = 1:length(u)
if abs(u(i)) > 1e-10
E(i) = sin(u(i))/u(i);
end
end

% Normalize
E_norm = abs(E)/max(abs(E));

% Convert to dB
E_dB = 20*log10(E_norm);
E_dB(E_dB < -40) = -40;

%% Cartesian Plot
figure;
plot(theta,E_norm,'LineWidth',2);
grid on;
xlabel('Angle (degrees)');
ylabel('Normalized Field');
```

```
title('H-Plane Radiation Pattern of Horn Antenna');
```

```
%% Polar Plot
```

```
figure;
```

```
polarplot(theta_rad,E_norm,'LineWidth',2);
```

```
title('Polar Radiation Pattern - H Plane');
```

```
%% dB Plot
```

```
figure;
```

```
plot(theta,E_dB,'r','LineWidth',2);
```

```
grid on;
```

```
xlabel('Angle (degrees)');
```

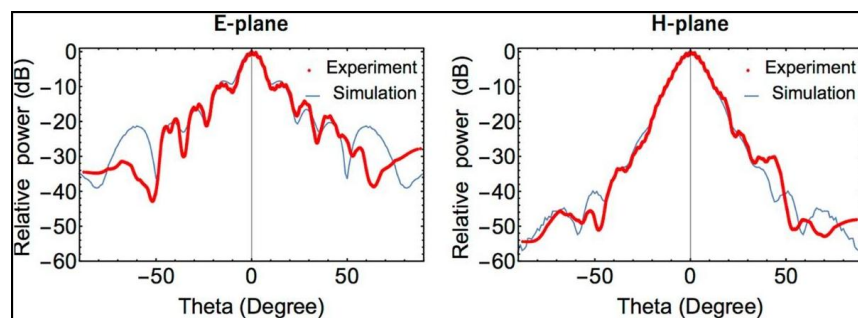
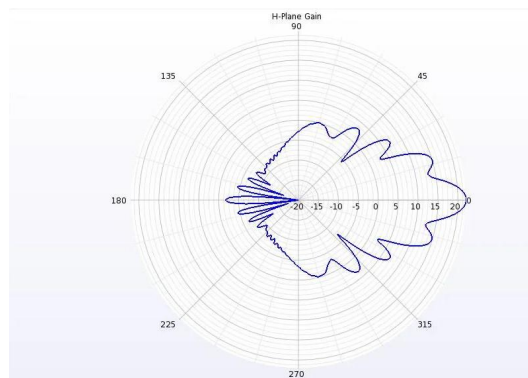
```
ylabel('Magnitude (dB)');
```

```
title('H-Plane Radiation Pattern (dB)');
```

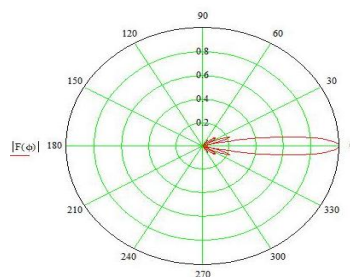
```
ylim([-40 0]);
```

Model Graph

- Polar radiation pattern.



- Radiation plot.



Tabular Column

Angle (°)	Gain (dB)
0	0 dB
±10	-1 dB
±20	-5 dB
±30	-13 dB
±45	-25 dB
±60	-35 dB

Example (Just for Understanding)

Half Power Beamwidth (HPBW)

For aperture width:

$$a = 6\lambda$$

Approximate HPBW:

$$HPBW \approx \frac{50.8\lambda}{a}$$

$$HPBW \approx \frac{50.8}{6}$$

$$HPBW \approx 8.5^\circ$$

Result

The MATLAB simulation successfully generated the H-plane radiation pattern of a microwave horn antenna. The antenna exhibits a highly directional main lobe with reduced beamwidth and low sidelobe levels, confirming the expected behavior of an H-plane horn antenna used in microwave communication systems.

Question 1

Simulation and Analysis of the Radiation Pattern of a Microwave E-plane Horn Antenna Using MATLAB

A microwave H-plane horn antenna is simulated using MATLAB to study its radiation pattern and field distribution characteristics at different operating frequencies.

Case 1

- Operating Frequency = 8 GHz
- Aperture Width = 6 cm

- Aperture Height = 4 cm

Case 2

- Operating Frequency = 10 GHz
- Aperture Width = 8 cm
- Aperture Height = 5 cm

Case 3

- Operating Frequency = 12 GHz
- Aperture Width = 10 cm
- Aperture Height = 6 cm

Tasks

1. Develop a MATLAB model of the horn antenna for the given operating conditions.
2. Simulate the radiation pattern of the horn antenna.
3. Plot the polar and 3D radiation patterns.
4. Determine the main lobe direction and radiation intensity.
5. Analyze the field distribution across the antenna aperture.
6. Compare the radiation characteristics for all three cases.
7. Identify the horn antenna configuration providing the most directional radiation pattern.

Question 2

Evaluation of Gain, Directivity, Beamwidth, and Radiation Efficiency of a Microwave H-plane Horn Antenna Using MATLAB

A H-plane horn antenna is analyzed under different operating conditions to evaluate its performance in terms of gain, directivity, beamwidth, and radiation efficiency.

Case 1

- Frequency = 8 GHz
- Gain = 14 dBi
- Beamwidth = 32°
- Radiation Efficiency = 85%

Case 2

- Frequency = 10 GHz
- Gain = 17 dBi
- Beamwidth = 24°
- Radiation Efficiency = 89%

Case 3

- Frequency = 12 GHz
- Gain = 20 dBi
- Beamwidth = 18°

- Radiation Efficiency = 92%

Tasks

1. Simulate the horn antenna using MATLAB.
2. Determine the gain, directivity, beamwidth, and radiation efficiency for each case.
3. Plot gain versus operating frequency.
4. Plot beamwidth versus operating frequency.
5. Compare the performance parameters of the horn antenna for all three cases.
6. Determine the operating condition providing maximum gain and directivity.
7. Analyze the relationship between beamwidth and antenna gain.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

EX NO: 10.1	Resonant Frequency Calculation using MATLAB
DATE:	

Objective

i) To calculate and simulate the resonant frequency for the mode $m = 1, n = 1$ using MATLAB based on the given resonant cavity or waveguide parameters.

(ii) To determine the resonant frequency for the mode $m = 3, n = 1$ using MATLAB by applying the appropriate resonant cavity or waveguide equations based on the given physical dimensions.

I) To calculate and simulate the resonant frequency for the mode $m=1,n=1$ using MATLAB based on the given resonant cavity or waveguide parameters.

Theory

In a rectangular waveguide, the resonant frequency for the TE mn mode is given by:

$$f_r = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

where:

- f_r = Resonant frequency (Hz)
- $c = 3 \times 10^8$ m/s (speed of light)
- a = Broad wall dimension (m)
- b = Narrow wall dimension (m)
- m, n = Mode numbers

For a WR-90 waveguide:

- $a = 22.86\text{mm} = 0.02286$ m
- $b = 10.16\text{mm} = 0.01016$ m

For TE $_{11}$:

- $m = 1$
- $n = 1$

Simulation Requirements

Software:

- MATLAB R2020a or later

Components Required:

- MATLAB Software
- Rectangular Waveguide Dimensions
- Mode Numbers (m,n)

SIMULATION PROCEDURE

1. Open MATLAB.
2. Define the waveguide dimensions.
3. Enter mode numbers $m = 1$, $n = 1$.
4. Apply the resonant frequency formula.
5. Calculate the resonant frequency.
6. Display the result.
7. Analyze the mode behavior.

MATLAB Code

```
clc;
clear;
close all;

c = 3e8;

a = 22.86e-3;
b = 10.16e-3;

m = 1;
n = 1;

fr = (c/2)*sqrt((m/a)^2 + (n/b)^2);

fprintf('Resonant Frequency = %.3f GHz\n',fr/1e9);
```

Observation Table

Parameter	Value
Waveguide Width (a)	22.86 mm
Waveguide Height (b)	10.16 mm

Mode Number (m)	1
Mode Number (n)	1
Parameter	Value

Simulation Output

Resonant Frequency = 16.150 GHz

Result

The resonant frequency for the TE₁₁ mode of the rectangular waveguide was successfully calculated using MATLAB and found to be approximately 16.15 GHz.

Objective(ii) To determine the resonant frequency for the mode $m = 3$, $n = 1$ using MATLAB by applying the appropriate resonant cavity or waveguide equations based on the given physical dimensions. This helps in understanding higher-order mode behavior in microwave resonant structures.

THEORETICAL BACKGROUND

The resonant frequency of a rectangular waveguide is:

$$f_r = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

For TE₃₁ mode:

- $m = 3$
- $n = 1$

The higher mode number increases the resonant frequency because more field variations occur within the waveguide.

Simulation Requirements

Software:

- MATLAB R2020a or later

Components Required:

- MATLAB Software
- WR-90 Waveguide Dimensions
- Mode Numbers ($m = 3$, $n = 1$)

Procedure

1. Open MATLAB.
2. Enter the waveguide dimensions.
3. Set $m = 3$ and $n = 1$.
4. Calculate the resonant frequency.
5. Display the obtained result.
6. Compare with the TE₁₁ mode frequency.

MATLAB CODE

```
clc;
clear;
close all;

c = 3e8;

a = 22.86e-3;
b = 10.16e-3;

m = 3;
n = 1;

fr = (c/2)*sqrt((m/a)^2 + (n/b)^2);

fprintf('Resonant Frequency = %.3f GHz\n',fr/1e9);
```

Observation Table

Parameter	Value
Waveguide Width (a)	22.86 mm
Waveguide Height (b)	10.16 mm
Mode Number (m)	3
Mode Number (n)	1
Resonant Frequency	25.09 GHz

Simulation Output

Resonant Frequency = 25.090 GHz

Result

The resonant frequency for the TE₃₁ mode of the rectangular waveguide was successfully calculated using MATLAB and found to be approximately 25.09 GHz. The result confirms that higher-order modes have higher resonant frequencies than lower-order modes.

Question 1: A rectangular resonant cavity is analyzed using MATLAB to determine the resonant frequency corresponding to the dominant mode ($m = 1$, $n = 1$). The cavity dimensions are varied to study their effect on the resonant frequency and electromagnetic field distribution.

Case 1

- Cavity Length (a) = 4 cm, Cavity Width (b) = 2 cm, Mode Numbers: $m = 1$, $n = 1$

Case 2

- Cavity Length (a) = 5 cm, Cavity Width (b) = 3 cm, Mode Numbers: $m = 1$, $n = 1$

Case 3 Cavity Length (a) = 6 cm

- Cavity Width (b) = 4 cm, Mode Numbers: $m = 1$, $n = 1$

Tasks

1. Develop a MATLAB program to calculate the resonant frequency for the mode $m=1$, $n=1$.
2. Determine the resonant frequency corresponding to each cavity dimension.
3. Tabulate the calculated resonant frequencies for all three cases.
4. Plot the variation of resonant frequency with cavity dimensions.
5. Compare the resonant frequencies obtained for the three cases.
6. Analyze the effect of cavity dimensions on the resonant frequency.

Question 2: A rectangular resonant cavity is analyzed using MATLAB to determine the resonant frequency corresponding to the higher-order mode $m=3$, $n=1$. The cavity dimensions are varied to investigate the effect of higher-order modes on the resonant characteristics of the structure.

Case 1

- Cavity Length (a) = 4 cm
- Cavity Width (b) = 2 cm
- Mode Numbers: $m = 3$, $n = 1$

Case 2

- Cavity Length (a) = 5 cm

- Cavity Width (b) = 3 cm
- Mode Numbers: m = 3, n = 1

Case 3

- Cavity Length (a) = 6 cm ,Cavity Width (b) = 4 cm,Mode Numbers: m = 3, n = 1

Tasks

1. Develop a MATLAB program to calculate the resonant frequency for the mode $m=3, n=1$.
2. Determine the resonant frequency corresponding to each cavity dimension.
3. Compare the resonant frequencies obtained for the $m=3, n=1$ mode with those of the dominant mode $m=1, n=1$.
4. Plot the variation of resonant frequency with cavity dimensions.
5. Analyze the influence of higher-order modes on the resonant behavior of the cavity.
6. Tabulate the resonant frequencies obtained for all three cases.
7. Identify the cavity configuration that produces the highest resonant frequency and explain the reason based on electromagnetic wave propagation principles.

Aim (10)	Tabular Column/ Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)